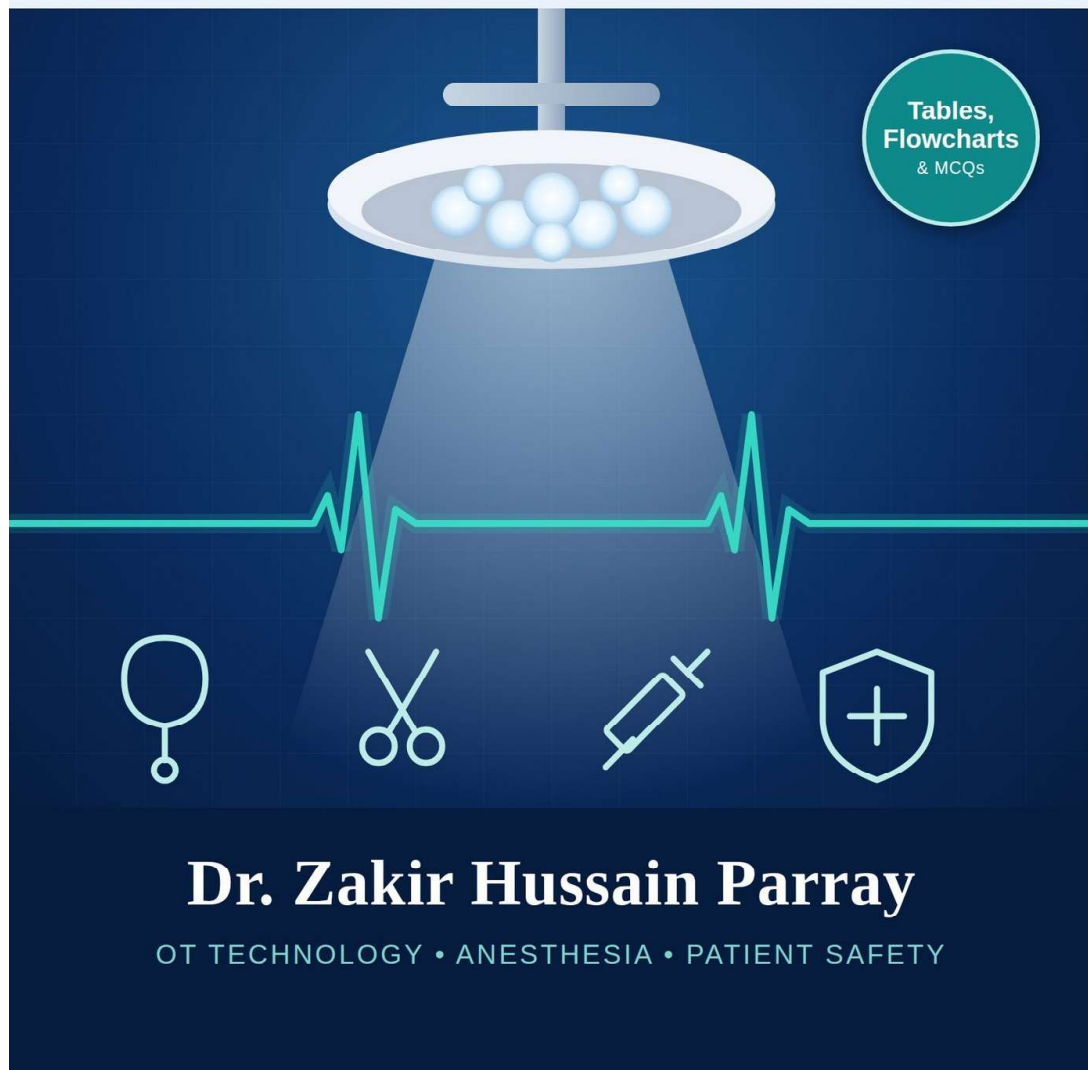


A COMPREHENSIVE TEXTBOOK

ANESTHESIA & OPERATION THEATRE TECHNOLOGY

A Basic Perspective

For Diploma, B.Sc. & B.Voc. Operation Theatre Technology Students



Anesthesia & Operation Theatre Technology

A Basic Perspective

Dr. Zakir Hussain Parray

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Anesthesia & Operation Theatre Technology: A Basic Perspective

First Edition, 2026

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All flowcharts, tables and illustrations in this book were created by the author for this edition.

Dedication

To every Operation Theatre Technology student who stands quietly beside the operating table at two in the morning — checking the suction, counting the swabs, watching the monitor — so that a stranger can wake up safely.

And to my teachers, who showed me that safety is not a rule to be followed but a habit to be lived.

Preface

The operation theatre is one of the most demanding workplaces in any hospital. Within a few square metres, a surgeon, an anesthesiologist, nurses and technicians must work together as a single unit while the patient — unconscious and completely dependent — trusts them with his or her life. The Operation Theatre (OT) and Anesthesia Technologist is the member of this team who keeps the environment, the equipment and the systems ready, safe and working. When the technologist does the job well, nobody notices. When something is missed, everyone notices.

This book was written because students of Diploma, B.Sc. and B.Voc. programmes in Operation Theatre and Anesthesia Technology have long needed a single, readable textbook that speaks to *them* — not to surgeons or to postgraduate anesthesiologists. Most available references are either pocket procedure lists written for another country's nursing system or heavy anesthesia texts that assume years of medical training. Students end up memorising scattered notes without understanding the “why” behind each practice.

Anesthesia & Operation Theatre Technology: A Basic Perspective tries to fill that gap. It starts from first principles — the structure of the OT, the anatomy that matters, the science of sterilisation — and builds step by step towards the anesthesia workstation, airway management, monitoring, drugs, regional anesthesia, perioperative care and emergencies. The final part covers the common surgical procedures a student will assist in, and the professional, legal and ethical responsibilities that come with the role.

Throughout the book I have tried to follow three principles:

1. **Clarity before detail.** Every chapter begins with the basic idea and then adds depth. A first-year diploma student should be able to follow it, and a final-year

B.Sc. student should still find enough detail to prepare for examinations and practical viva.

2. **Visual learning.** Operation theatre work is procedural. Flowcharts show *sequences*; tables show *comparisons*. Wherever a concept can be shown instead of described, I have used a table or flowchart.
3. **Safety first.** Every chapter contains *Safety Alert* boxes that highlight the errors that actually harm patients in real theatres — wrong-site surgery, retained swabs, oxygen pipeline cross-connection, drug errors, fires, and burns.

I hope this book becomes a companion that students carry from their first day in the OT to their final examinations, and beyond into their working lives. Suggestions from teachers and students for improving future editions will be gratefully received.

Dr. Zakir Hussain Parray

How to Use This Book

Each chapter follows the same structure so that you always know where to look.

Feature	What it contains	How to use it
Learning Objectives	What you should be able to do after the chapter	Read before the chapter; test yourself after
Key Terms	Important words defined briefly	Learn these first — they are common viva questions
Tables	Comparisons, classifications, doses, checklists	Ideal for last-minute revision
Flowcharts	Step-by-step sequences and decision pathways	Practise drawing them — they score marks in examinations
Clinical Pearl (green box)	Practical tips from real theatre experience	Apply them during your clinical postings
Safety Alert (red box)	Errors that can injure patients or staff	Never ignore — these are “never events”
Remember (purple box)	Mnemonics and memory aids	Use for quick recall
Chapter Summary (yellow box)	Key points in brief	Revise the night before an exam
Review Questions	MCQs with answers, short and long questions	Self-assessment and exam preparation

Clinical Pearl — Study tip: Read a chapter *before* your clinical posting on that topic, observe the practice in the theatre, and then read the chapter again. Knowledge that is seen, done and then re-read is rarely forgotten.

A note on terminology

- **Anesthesia / anaesthesia:** Both spellings are correct. This book uses the American spelling *anesthesia* (as in the title), except in the names of organisations that use the British form.
- **OT, OR and theatre** all mean the operating room.
- **OT Technician / OT Technologist / Anesthesia Technologist / Surgical Technologist** are used for the same allied health professional; the exact title differs between institutions and states.
- Drug doses are given for **healthy adults** unless stated otherwise. Doses for children are usually calculated per kilogram of body weight.

Common abbreviations used in this book

Abbreviation	Meaning	Abbreviation	Meaning
ASA	American Society of Anesthesiologists	MAC	Minimum Alveolar Concentration
BMW	Bio-Medical Waste	NIBP	Non-Invasive Blood Pressure
CSSD	Central Sterile Supply Department	NMB	Neuromuscular Blocker
ETT	Endotracheal Tube	PACU	Post-Anesthesia Care Unit
ETCO ₂	End-Tidal Carbon Dioxide	PPE	Personal Protective Equipment
FGF	Fresh Gas Flow	SpO ₂	Peripheral Oxygen Saturation
LMA	Laryngeal Mask Airway	SSI	Surgical Site Infection
LA	Local Anesthetic	TIVA	Total Intravenous Anesthesia

A complete list of abbreviations is given in **Appendix A**.

Part I — Foundations

Part I lays the groundwork for everything that follows. It introduces the profession of the Operation Theatre and Anesthesia Technologist, reviews the anatomy and physiology you need in daily practice, and describes how a modern operation theatre complex is designed to keep patients safe.

- Chapter 1: Introduction to Operation Theatre & Anesthesia Technology
- Chapter 2: Applied Anatomy & Physiology for the OT Technologist
- Chapter 3: The Operation Theatre Complex — Design, Zones & Environment

Chapter 1: Introduction to Operation Theatre & Anesthesia Technology

Learning Objectives — After studying this chapter, you should be able to:

- Define Operation Theatre Technology and Anesthesia Technology.
- Outline the historical milestones that shaped modern surgery and anesthesia.
- List the members of the surgical team and describe each person's role.
- Explain the duties of the OT technologist before, during and after surgery.
- Distinguish between *sterile* and *non-sterile* team members.
- Describe the personal qualities, ethics and communication skills required of an OT professional.
- Outline the education pathway and career opportunities in the field.

Key Terms: Perioperative · Scrub person · Circulator · Anesthesia technologist · Sterile field · Surgical conscience · Asepsis · Scope of practice · Standard Operating Procedure (SOP)

1.1 What Is Operation Theatre Technology?

Operation Theatre (OT) Technology is the branch of allied health science concerned with preparing, maintaining and managing the environment, equipment, instruments and supplies needed to perform surgical and anesthetic procedures safely. The OT technologist works under the direction of the surgeon and anesthesiologist and alongside nursing staff to ensure that every operation is carried out in a safe, clean and efficient setting.

Anesthesia Technology is the closely related discipline that focuses on the preparation, checking, operation and maintenance of anesthesia equipment — the anesthesia workstation, airway devices, monitors, infusion pumps and emergency equipment — and on assisting the anesthesiologist during the induction, maintenance and recovery phases of anesthesia.

In most Indian institutions, the two roles are taught together in a combined programme usually called **Operation Theatre and Anesthesia Technology (OTAT or AOTT)**. The graduate may work in either role depending on the needs of the hospital.

The word **perioperative** describes the entire period surrounding surgery. It has three phases, and the OT technologist has responsibilities in all three.

Phase	Begins	Ends	Main focus for the technologist
Preoperative	Decision to operate	Patient transferred to the OT table	Room and equipment preparation, patient identification, checklist
Intraoperative	Patient placed on the OT table	Patient admitted to the recovery room (PACU)	Scrubbing or circulating, assisting anesthesia, counts, safety
Postoperative	Admission to PACU	Patient fully recovered from the effects of surgery and anesthesia	Transfer, recovery monitoring, cleaning and turnover of the OT

1.2 A Brief History of Surgery and Anesthesia

For most of human history, surgery was a terrifying last resort. Operations were performed without pain relief, and patients who survived the shock of the procedure frequently

died of infection afterwards. Three great discoveries of the nineteenth century — **anesthesia**, **antisepsis** and later **asepsis** — transformed surgery into the safe, planned discipline we know today.

1.2.1 Ancient roots

Indian medicine has a proud surgical tradition. The *Sushruta Samhita*, compiled more than two thousand years ago, describes over a hundred surgical instruments, techniques of reconstructive surgery of the nose, cataract couching, and the use of wine and herbal preparations to dull pain. Sushruta is widely regarded as the “Father of Surgery”. Ancient Egyptian, Greek and Chinese physicians also practised surgery, but pain and infection limited what could be achieved.

1.2.2 Milestones in the development of modern surgery and anesthesia

Year	Milestone	Significance
c. 600 BCE	<i>Sushruta Samhita</i> (India)	Detailed descriptions of surgical instruments and procedures
1772	Joseph Priestley discovers nitrous oxide	Later used as the first anesthetic gas
1844	Horace Wells uses nitrous oxide for dental extraction	Early attempt at surgical pain relief
16 October 1846	William T. G. Morton demonstrates ether anesthesia in Boston	Celebrated worldwide as World Anaesthesia Day
1847	James Young Simpson introduces chloroform	Popularised anesthesia in childbirth
1867	Joseph Lister publishes on antiseptic surgery using carbolic acid	Dramatic reduction in post-operative infection
1884	Carl Koller uses cocaine as a topical anesthetic for the eye	Birth of local anesthesia

Year	Milestone	Significance
1890	William Halsted introduces rubber gloves in the OT	Foundation of barrier protection
1898	August Bier performs the first planned spinal anesthesia	Beginning of regional anesthesia
1942	Griffith and Johnson use curare as a muscle relaxant	Allowed lighter planes of general anesthesia
1956	Halothane introduced	First modern halogenated volatile agent
1981	Archie Brain invents the laryngeal mask airway	Revolutionised airway management
1987 onwards	Laparoscopic cholecystectomy	Era of minimal access surgery
2008	WHO Surgical Safety Checklist launched	Global standard for safe surgery

Remember — “EAR” of modern surgery: Ether (pain control, 1846) → Antisepsis (infection control, 1867) → Rubber gloves & asepsis (barrier protection, 1890s).

1.3 The Surgical Team

Surgery is a team activity. Each member has a defined role, and safe surgery depends on every member performing that role and communicating clearly with the others. The team is divided into two groups based on whether they touch the **sterile field**.

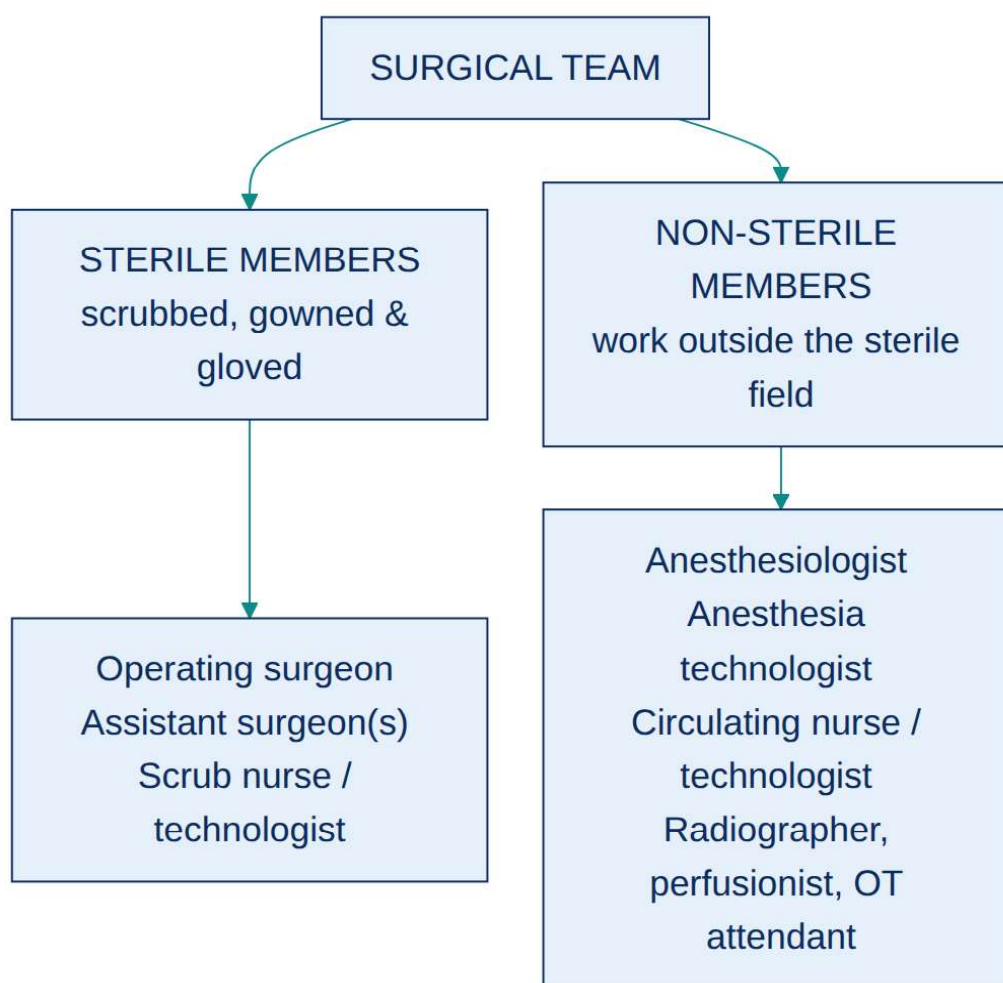


Figure 1.1 — Members of the surgical team

1.3.1 Roles of each team member

Team member	Sterile / Non-sterile	Principal responsibilities
Operating surgeon	Sterile	Leads the surgical procedure; responsible for diagnosis, surgical plan, consent and the operation itself
Assistant surgeon	Sterile	Provides exposure (retraction), haemostasis, suction and suturing under the surgeon's direction
Scrub technologist	Sterile	Sets up the sterile

Team member	Sterile / Non-sterile	Principal responsibilities
/ nurse		instrument trolley, passes instruments, maintains sterility, performs counts with the circulator
Anesthesiologist	Non-sterile	Pre-anesthetic assessment, administration of anesthesia, monitoring and resuscitation, pain management
Anesthesia technologist	Non-sterile	Checks the anesthesia machine and equipment, prepares drugs and airway devices, assists in induction, intubation, monitoring and emergencies
Circulating technologist / nurse	Non-sterile	Manages the room, opens sterile supplies, brings additional items, documents counts and events, positions the patient, handles specimens
OT attendant / orderly	Non-sterile	Patient transfer, cleaning, moving equipment, transport of specimens and supplies
Other specialists	Usually non-sterile	Radiographer (C-arm), perfusionist (heart-lung machine), biomedical engineer, pathologist for frozen sections

Clinical Pearl: A good scrub technologist *anticipates* the next

instrument. Learn the steps of each operation (Chapter 19) so that the right instrument is ready before the surgeon asks for it. Anticipation shortens operating time and reduces anesthesia exposure for the patient.

1.4 Duties of the OT Technologist

The duties of the OT technologist can be grouped according to the three phases of perioperative care.

1.4.1 Before surgery (preoperative duties)

- Checking the operating list and preparing the theatre for each case.
- Damp-dusting all horizontal surfaces, lights and equipment at the start of the day.
- Checking the OT table, lights, suction, electrosurgical unit, and gas supplies.
- Collecting the correct instrument sets, sutures, drapes and special equipment.
- Checking the sterility indicators and expiry dates of all packs.
- Assisting the anesthesia team to check the anesthesia machine, monitors and airway equipment.
- Receiving the patient, verifying identity, consent, site marking, fasting status and investigations.

1.4.2 During surgery (intraoperative duties)

As the scrub technologist:

- Performing surgical hand antisepsis, gowning and gloving.
- Setting up the instrument trolley and Mayo stand in a standard order.
- Performing the initial count of swabs, sharps and instruments with the circulator.
- Assisting with draping and passing instruments in the correct manner.
- Maintaining the integrity of the sterile field throughout.
- Receiving and handling specimens correctly.

- Performing closing counts before the cavity and skin are closed.

As the circulating technologist:

- Positioning the patient safely with the surgeon and anesthesiologist.
- Performing or assisting with skin preparation.
- Connecting diathermy, suction, light sources and other equipment.
- Opening sterile items onto the field using aseptic technique.
- Recording counts, times, implants and specimens.
- Monitoring the room for breaks in asepsis and maintaining traffic control.

As the anesthesia technologist:

- Preparing drugs, infusions and IV fluids as directed.
- Assisting with pre-oxygenation, induction, intubation and securing of the airway.
- Attaching and troubleshooting monitors.
- Maintaining records of vital signs and fluids where required.
- Being ready with the difficult airway trolley and emergency drugs.

1.4.3 After surgery (postoperative duties)

- Assisting with extubation and transfer of the patient to the PACU.
- Handing over the patient with relevant information.
- Disposing of sharps, waste and linen according to biomedical waste rules.
- Decontaminating and sending instruments to the CSSD.
- Cleaning and preparing the room for the next case (turnover).
- Cleaning, disinfecting and restocking the anesthesia machine and circuits.
- Terminal cleaning at the end of the day.

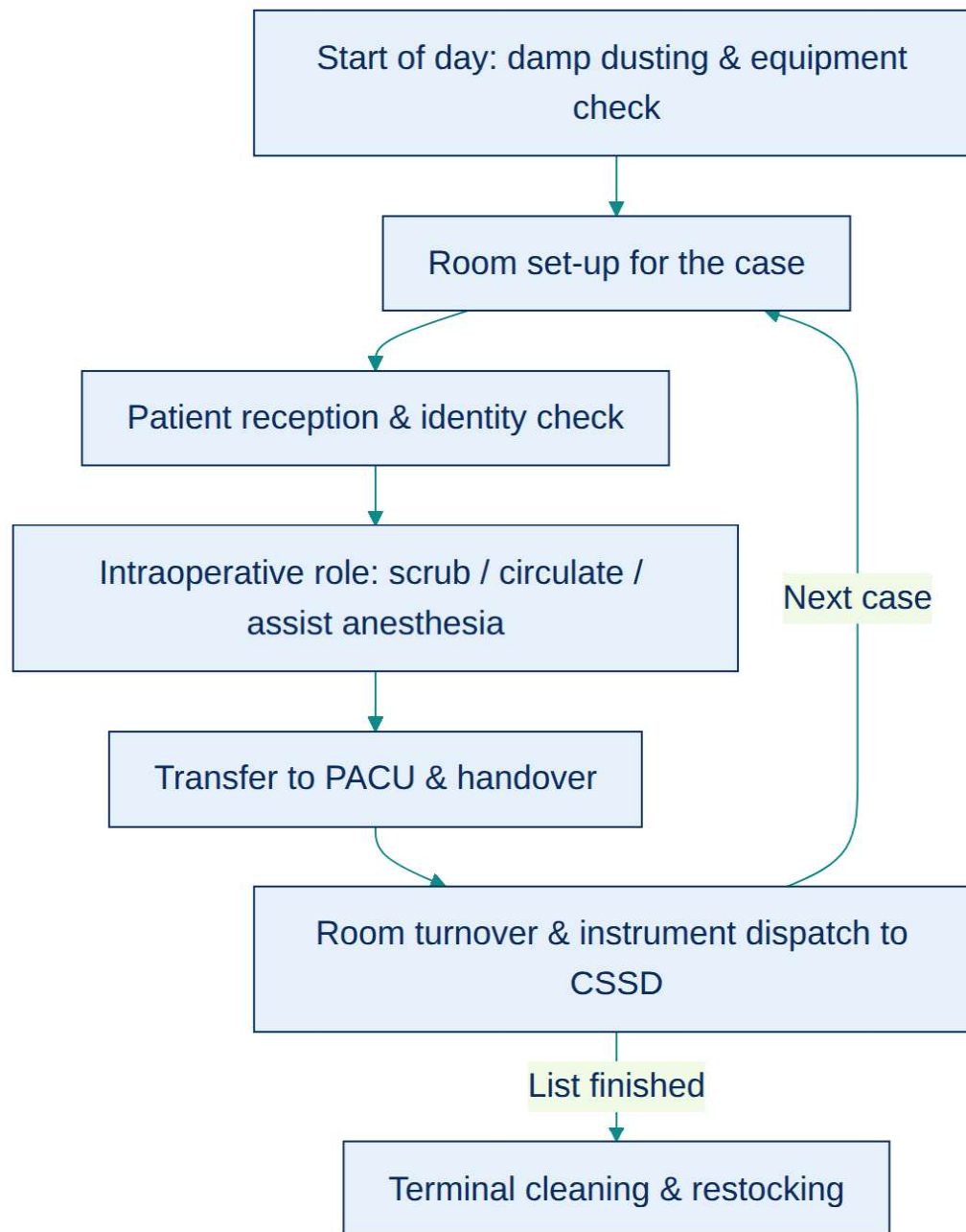


Figure 1.2 — The daily workflow of an OT technologist

1.5 Qualities of a Good OT Professional

Technical knowledge is only part of what makes an effective OT technologist. The following qualities are equally important.

Quality	Why it matters in the OT
Surgical conscience	The personal commitment to admit and correct any break in sterile technique, even if no one else saw it
Alertness and anticipation	Emergencies can develop within seconds; the alert technologist notices changes early
Physical stamina	Long hours of standing, lifting and wearing PPE
Emotional stability	Remaining calm during bleeding, cardiac arrest or conflict
Attention to detail	Counts, labels, expiry dates and drug names must be exact
Teamwork and communication	Clear closed-loop communication prevents errors
Manual dexterity	Handling fine instruments, loading needles, assembling equipment
Honesty and integrity	Accurate documentation and reporting of errors
Respect for patient dignity	The patient is anaesthetised and vulnerable; privacy must be protected at all times
Willingness to learn	Technology in the OT changes rapidly

Safety Alert — Surgical conscience: If you touch a non-sterile surface with a sterile glove, *say so immediately* and change the glove. Hiding a contamination to avoid embarrassment can lead to a surgical site infection, a second operation or even death. No professional will think less of you for being honest; they will think less of you for hiding it.

1.6 Communication in the Operation Theatre

Most errors in healthcare are caused not by a lack of knowledge but by failures of communication. Several structured communication tools are used in theatres.

1.6.1 Closed-loop communication

The sender gives a clear instruction, the receiver repeats it back, and the sender confirms. For example:

Anesthesiologist: "Please draw up 100 micrograms of fentanyl." **Technologist:** "Drawing up 100 micrograms of fentanyl."
Anesthesiologist: "Correct."

1.6.2 SBAR handover

SBAR is a structured method for handing over a patient or reporting a problem.

Letter	Stands for	Example (handover to PACU)
S	Situation	"This is Mr. Kumar, 45 years, after laparoscopic cholecystectomy under general anesthesia."
B	Background	"He is hypertensive on amlodipine; no allergies."
A	Assessment	"Extubated awake, SpO ₂ 97% on 4 L oxygen, BP 130/80, pain score 3."
R	Recommendation	"Continue oxygen, monitor for nausea, paracetamol due at 6 pm."

1.6.3 Time-out and briefings

The WHO Surgical Safety Checklist (Chapter 8) formalises communication at three points: before induction (**Sign In**), before incision (**Time Out**) and before the patient leaves the theatre (**Sign Out**). Many hospitals also hold a brief team meeting at the start of the list (**briefing**) and at its end (**debriefing**).

1.7 Ethics and Professional Conduct

OT technologists must follow the basic principles of healthcare ethics.

Principle	Meaning	Application in the OT
Autonomy	Respect for the patient's right to decide	Valid informed consent before surgery
Beneficence	Acting in the patient's best interest	Meticulous preparation and monitoring
Non-maleficence	"First, do no harm"	Correct positioning, counts, asepsis
Justice	Fair and equal treatment	Same standard of care for all patients regardless of status
Confidentiality	Protecting patient information	No photographs, no discussion of patients outside the OT
Veracity	Truthfulness	Honest documentation and reporting of incidents

Safety Alert — Mobile phones and photographs: Never photograph a patient, an operation or the patient's records on your personal phone, and never share such images on social media. This is a serious breach of confidentiality and may lead to disciplinary and legal action.

1.8 Scope of Practice

The **scope of practice** defines what an OT technologist is trained and permitted to do. It varies between institutions and countries, but in general:

The OT technologist MAY (under supervision)	The OT technologist must NOT
Prepare the theatre, instruments and equipment	Administer anesthesia independently

The OT technologist MAY (under supervision)	The OT technologist must NOT
Act as scrub person or circulator	Obtain consent for surgery
Assist in intubation, cannulation and monitoring	Prescribe medicines
Prepare drugs as directed and labelled	Perform any surgical step independently unless specifically trained and authorised
Assist in resuscitation (BLS)	Alter a doctor's instructions
Operate and troubleshoot equipment	Work beyond competence or without supervision where required

In India, allied health professions are regulated under the **National Commission for Allied and Healthcare Professions (NCAHP) Act, 2021**, which aims to standardise education and registration for professions including Operation Theatre and Anesthesia Technology.

1.9 Education and Career Pathways

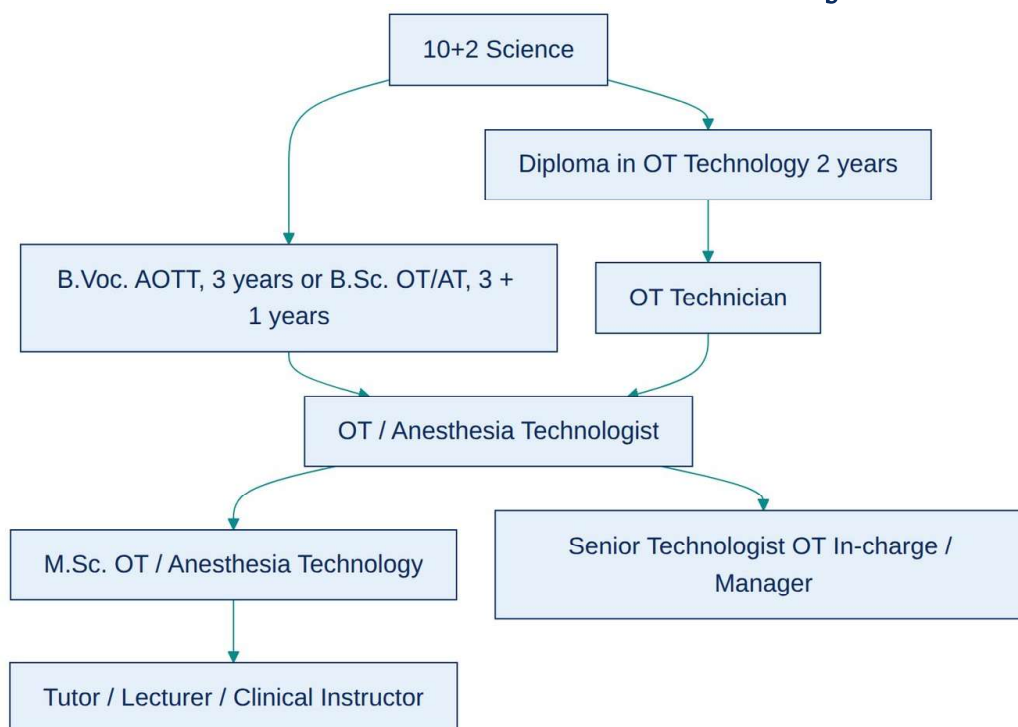


Figure 1.3 — Education and career pathway in OT and Anesthesia Technology

Employment opportunities exist in government and private hospitals, multispeciality and super-speciality centres, day-care surgical centres, trauma and emergency units, ICUs, CSSDs, medical equipment companies and teaching institutions. Opportunities also exist abroad for qualified and registered professionals.

1.10 Standard Operating Procedures (SOPs)

A **Standard Operating Procedure** is a written, step-by-step instruction for performing a routine task in a uniform way. SOPs reduce variation between staff, help in training new members and provide a reference during audits and accreditation (for example, by the **National Accreditation Board for Hospitals & Healthcare Providers, NABH**).

Typical OT SOPs include:

- Daily OT preparation and cleaning.
- Anesthesia machine checking.
- Surgical count procedure.
- Handling of specimens.
- Biomedical waste segregation.
- Fire safety and evacuation.
- Management of a needle-stick injury.
- Cleaning and sterilisation of flexible endoscopes.

Chapter Summary

- OT Technology prepares and manages the surgical environment; Anesthesia Technology supports the anesthesiologist with equipment and assistance.
- The perioperative period has three phases — preoperative, intraoperative and postoperative.
- Ether (1846), antisepsis (1867) and gloves/asepsis (1890s) made modern surgery possible.
- The surgical team has sterile members (surgeon, assistant, scrub person) and non-sterile members (anesthesiologist, anesthesia technologist, circulator, support staff).
- Surgical conscience, alertness and clear communication are the most important professional qualities.
- Closed-loop communication, SBAR and the WHO

checklist are tools to prevent errors.

- The technologist works within a defined scope of practice under supervision.

Review Questions

Multiple Choice Questions

1. World Anaesthesia Day is observed on:
a) 16 September b) 16 October c) 1 July d) 7 April
2. Who introduced antiseptic surgery using carbolic acid?
a) Halsted b) Morton c) Lister d) Simpson
3. Which of the following is a **non-sterile** member of the surgical team?
a) Assistant surgeon b) Scrub technologist c) Circulating technologist d) Operating surgeon
4. The first planned spinal anesthesia was performed by:
a) Carl Koller b) August Bier c) Archie Brain d) Horace Wells
5. The personal commitment to report any break in sterile technique is called:
a) Asepsis b) Surgical conscience c) Beneficence d) Veracity
6. In SBAR, the letter "A" stands for:
a) Action b) Airway c) Assessment d) Allergy
7. The laryngeal mask airway was invented by:
a) Archie Brain b) Macintosh c) Magill d) Miller
8. The perioperative intraoperative phase ends when:

- a) Skin is closed b) Anesthesia is stopped c) Patient is admitted to PACU d) Patient is discharged home

Answers: 1-b, 2-c, 3-c, 4-b, 5-b, 6-c, 7-a, 8-c

Short Answer Questions

1. Define perioperative care and list its three phases.
2. List the members of the surgical team and classify them as sterile or non-sterile.
3. What is surgical conscience? Give one example.
4. Explain closed-loop communication with an example.
5. List six qualities of a good OT technologist.

Long Answer Questions

1. Describe the duties of the OT technologist in the preoperative, intraoperative and postoperative phases.
2. Trace the history of anesthesia and surgery from ancient times to the introduction of the WHO Surgical Safety Checklist.
3. Discuss the ethical principles that guide the work of an OT professional, with examples from theatre practice.

Chapter 2: Applied Anatomy & Physiology for the OT Technologist

Learning Objectives — After studying this chapter, you should be able to:

- Describe the anatomy of the upper and lower airway relevant to intubation.
- Explain the mechanics of breathing, lung volumes and gas exchange.
- Describe the conduction system of the heart and the determinants of blood pressure.
- Outline the anatomy of the spinal cord and vertebral column relevant to spinal and epidural anesthesia.
- Explain the neuromuscular junction and how muscle relaxants act.
- Describe body fluid compartments and the normal values of common laboratory tests.
- Explain the physiology of temperature regulation and why patients become cold in the OT.

Key Terms: Larynx · Vocal cords · Carina · Tidal volume · Functional residual capacity (FRC) · Dead space · Cardiac output · Mean arterial pressure · Subarachnoid space · Ligamentum flavum · Neuromuscular junction · Acetylcholine · Hypothermia

2.1 Why Anatomy Matters to the OT Technologist

The anesthesia technologist handles the patient's airway, positions the patient, assists with spinal and epidural blocks, attaches monitors and prepares drugs that act on the heart, lungs and nerves. Every one of these tasks is safer when the technologist understands the structures involved. This chapter does not attempt to cover the whole

of human anatomy; it concentrates on the systems that matter most in the operation theatre.

2.2 The Respiratory System

2.2.1 The upper airway

The upper airway extends from the nose and mouth to the larynx. It warms, humidifies and filters inspired air.

Structure	Key features	Relevance in the OT
Nose (nasal cavity)	Divided by the nasal septum; lined by vascular mucosa; three turbinates on each lateral wall	Nasal intubation and nasopharyngeal airways can cause bleeding; use lubricant and vasoconstrictor drops
Oral cavity	Teeth, tongue, hard and soft palate	Loose teeth may be dislodged during laryngoscopy; the tongue is the commonest cause of airway obstruction in the unconscious patient
Pharynx	Muscular tube divided into nasopharynx, oropharynx and laryngopharynx (hypopharynx)	Oropharyngeal (Guedel) airway lifts the tongue off the posterior pharyngeal wall
Epiglottis	Leaf-shaped cartilage that covers the laryngeal inlet during swallowing	Macintosh blade tip sits in the vallecula (space in front of the epiglottis); Miller blade lifts the epiglottis directly
Larynx	Cartilaginous “voice box” at the level of C3-C6 in the adult	Contains the vocal cords; the narrowest part of the adult airway is the glottis (rima glottidis)

2.2.2 The larynx

The larynx is formed by nine cartilages — three single (thyroid, cricoid, epiglottis) and three paired (arytenoid, corniculate, cuneiform).

Cartilage	Description	Clinical importance
Thyroid	Largest; forms the “Adam’s apple”	Landmark for airway assessment and for the BURP manoeuvre (Backward, Upward, Rightward Pressure) to improve the laryngoscopic view
Cricoid	The only complete ring of cartilage in the airway	Target for cricoid pressure (Sellick manoeuvre) during rapid sequence induction; narrowest part of the airway in young children
Epiglottis	Elastic cartilage attached to the thyroid	Guides laryngoscope blade placement
Arytenoids	Pyramid-shaped; vocal cords attach to them	May be dislocated by traumatic intubation

The **cricothyroid membrane**, between the thyroid and cricoid cartilages, is the site for emergency front-of-neck access (cricothyroidotomy) when a patient can neither be intubated nor oxygenated.

Clinical Pearl — Adult versus child airway: A child’s airway is not simply a small adult airway. Infants have a relatively large head and tongue, a higher (C3–C4) and more anterior larynx, a floppy U-shaped epiglottis, and the narrowest point at the cricoid ring. A straight (Miller) blade and a shoulder roll are often helpful.

Feature	Infant	Adult
Level of larynx	C3–C4 (higher)	C4–C6
Epiglottis	Long, floppy, U-	Flat, firm

Feature	Infant	Adult
	shaped	
Narrowest part	Cricoid ring (subglottic)	Glottis (vocal cords)
Tongue	Relatively large	Proportionate
Occiput	Large — neck flexes when supine	Normal — a pillow is needed for sniffing position
Preferred blade	Straight (Miller)	Curved (Macintosh)
Breathing	Obligate nasal breathers (first months)	Nasal and oral

2.2.3 The lower airway

The **trachea** is a tube about 10–12 cm long and 2 cm wide in adults, supported by 16–20 C-shaped cartilaginous rings. It begins at C6 and divides at the **carina** (level of T4–T5, the sternal angle) into the right and left main bronchi.

Feature	Right main bronchus	Left main bronchus
Length	Shorter (about 2.5 cm)	Longer (about 5 cm)
Width	Wider	Narrower
Angle from trachea	More vertical (about 25°)	More horizontal (about 45°)
Clinical point	Endotracheal tube pushed too far usually enters the right bronchus; aspirated foreign bodies also go right	Double-lumen tubes are usually left-sided for easier placement

Safety Alert — Endobronchial intubation: If the ETT is advanced too far, only the right lung is ventilated. Signs are unequal chest movement, absent breath sounds on the left, high airway pressure and falling SpO₂. Typical correct depth at the teeth is about **21 cm for adult women and 23 cm for adult men**. Always auscultate both sides after intubation.

2.2.4 Mechanics of breathing

Air flows from an area of higher pressure to one of lower pressure. During **inspiration**, the diaphragm contracts and moves downward and the external intercostal muscles lift the ribs, enlarging the chest cavity. Pressure inside the lungs falls below atmospheric pressure, and air flows in. **Expiration** at rest is passive — the elastic lungs recoil.

When a patient is mechanically ventilated, this relationship is reversed: the ventilator *pushes* gas into the lungs with **positive pressure**. This has important effects on the circulation (reduced venous return and cardiac output) that the anesthesia team must remember.

2.2.5 Lung volumes and capacities

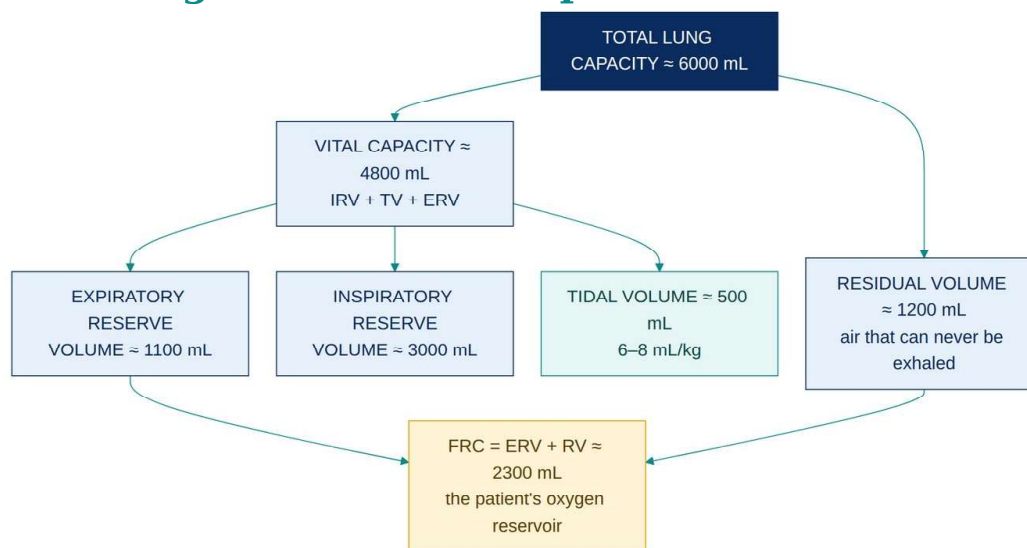


Figure 2.1 — Lung volumes and capacities (typical adult values)

Volume / capacity	Definition	Normal adult value
Tidal volume (TV)	Volume breathed in or out in a normal breath	500 mL (6–8 mL/kg)
Inspiratory reserve volume (IRV)	Extra volume that can be inhaled after a normal inspiration	~3000 mL
Expiratory reserve volume (ERV)	Extra volume that can be exhaled after a normal expiration	~1100 mL

Volume / capacity	Definition	Normal adult value
Residual volume (RV)	Volume remaining after maximal expiration	~1200 mL
Functional residual capacity (FRC)	ERV + RV — volume left in the lungs at the end of a normal breath	~2300 mL
Vital capacity (VC)	IRV + TV + ERV	~4800 mL
Total lung capacity (TLC)	VC + RV	~6000 mL
Minute ventilation	TV × respiratory rate	500 × 12 = 6 L/min
Anatomical dead space	Air in the conducting airways that does not take part in gas exchange	~150 mL (2 mL/kg)

Clinical Pearl — Why we pre-oxygenate: The FRC is the patient's oxygen store during apnoea. Breathing 100% oxygen for 3 minutes (or 8 deep breaths) replaces the nitrogen in the FRC with oxygen — called **denitrogenation**. A healthy adult can then remain apnoeic for several minutes before desaturating. FRC is reduced in obesity, pregnancy, the supine position and under general anesthesia, so these patients desaturate much faster.

2.2.6 Gas exchange and oxygen transport

Gas exchange takes place across the thin alveolar-capillary membrane by **diffusion**. Oxygen moves from the alveoli (high partial pressure) into the blood, and carbon dioxide moves in the opposite direction.

About 98% of oxygen in blood is carried bound to **haemoglobin**; only about 2% is dissolved in plasma. Each gram of haemoglobin, when fully saturated, can carry 1.34 mL of oxygen. The pulse oximeter measures the percentage of haemoglobin that is saturated with oxygen (**SpO₂**).

Parameter	Normal value (breathing room air)
PaO ₂ (arterial oxygen tension)	80-100 mmHg

Parameter	Normal value (breathing room air)
PaCO ₂ (arterial carbon dioxide tension)	35–45 mmHg
SaO ₂ / SpO ₂	95–100%
pH	7.35–7.45
HCO ₃ ⁻ (bicarbonate)	22–26 mEq/L
ETCO ₂	35–45 mmHg (usually 2–5 mmHg lower than PaCO ₂)

Remember — Oxygen-haemoglobin dissociation curve
“30-60-90”: a PaO₂ of 30, 60 and 90 mmHg corresponds approximately to saturations of 60%, 90% and near 97–100%. Below an SpO₂ of about 90%, oxygen levels fall steeply — the “cliff edge”.

2.3 The Cardiovascular System

2.3.1 Structure of the heart

The heart is a four-chambered muscular pump. The right side receives deoxygenated blood from the body and pumps it to the lungs (pulmonary circulation); the left side receives oxygenated blood from the lungs and pumps it to the body (systemic circulation).

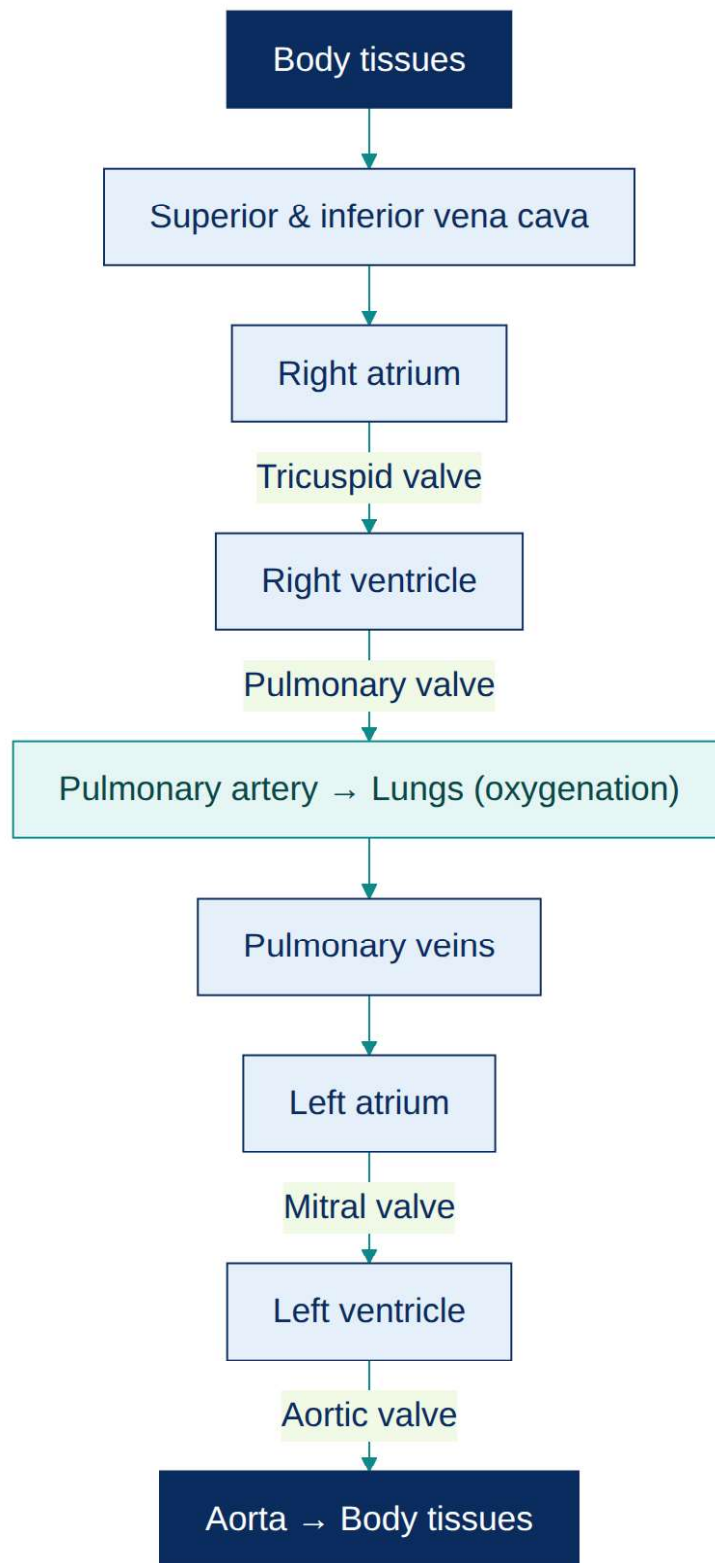


Figure 2.2 — Path of blood through the heart

2.3.2 Conduction system and the ECG

Each heartbeat begins with an electrical impulse generated in the **sinoatrial (SA) node**, the natural pacemaker, located in the right atrium. The impulse spreads through the atria, is briefly delayed at the **atrioventricular (AV) node**, then passes down the **bundle of His**, the right and left **bundle branches**, and the **Purkinje fibres** to the ventricles.

ECG component	Represents	Normal duration
P wave	Atrial depolarisation	< 0.12 s
PR interval	Time from atrial depolarisation to ventricular depolarisation (includes AV node delay)	0.12–0.20 s
QRS complex	Ventricular depolarisation	< 0.12 s
ST segment	Plateau between depolarisation and repolarisation	Isoelectric (flat)
T wave	Ventricular repolarisation	—
QT interval	Total ventricular activity	< 0.44 s (corrected for rate)

Heart rhythms and their monitoring are described in Chapter 12.

2.3.3 Blood pressure and cardiac output

Cardiac output (CO) is the volume of blood pumped by the heart each minute:

$$\text{Cardiac output} = \text{Heart rate} \times \text{Stroke volume}$$

$$\text{Example: } 70 \text{ beats/min} \times 70 \text{ mL} = 4.9 \text{ L/min}$$

Blood pressure (BP) depends on the cardiac output and the resistance of the blood vessels:

$$\text{Blood pressure} = \text{Cardiac output} \times \text{Systemic vascular resistance}$$

Mean arterial pressure (MAP) is the average pressure driving blood into the organs:

MAP $\approx$ Diastolic pressure + $\frac{1}{3}$ (Systolic – Diastolic) Example: BP 120/80 $\rightarrow$ MAP = 80 + $\frac{1}{3}(40) \approx 93$ mmHg

A MAP of at least **65 mmHg** is usually needed to perfuse the brain, heart and kidneys adequately.

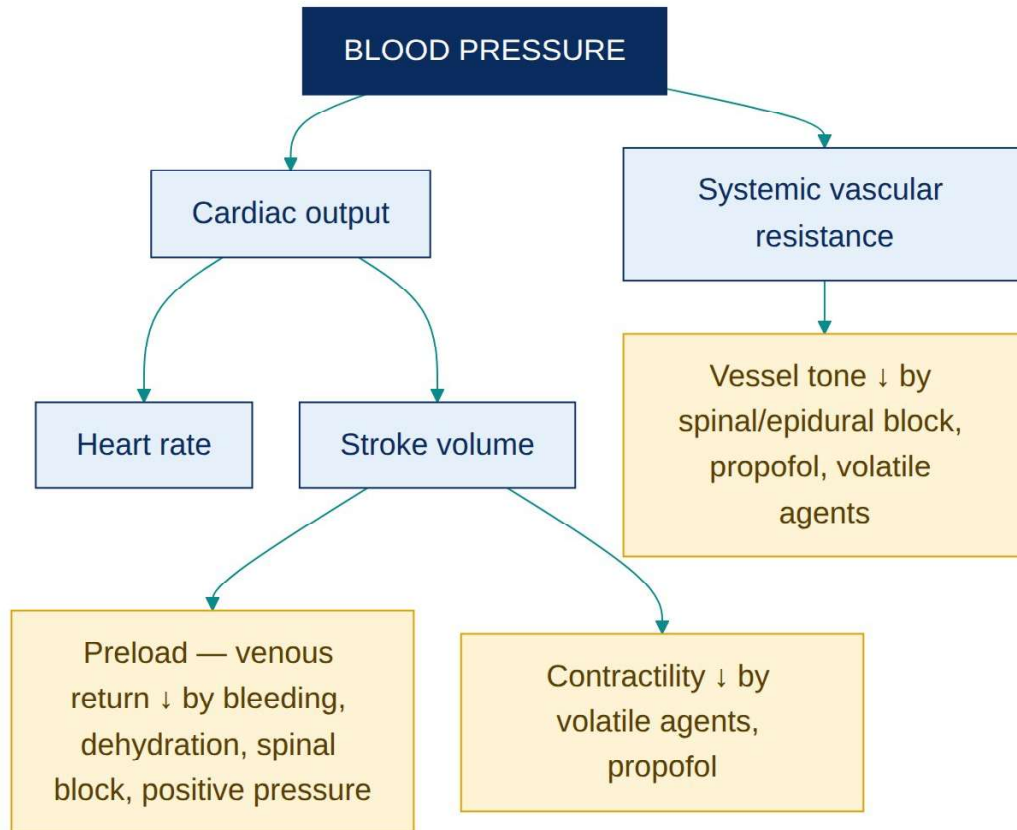


Figure 2.3 — Factors that determine blood pressure (and how anesthesia affects them)

Clinical Pearl: Almost every anesthetic technique lowers blood pressure — general anesthetic drugs reduce vessel tone and contractility, and spinal anesthesia blocks the sympathetic nerves that keep vessels constricted. This is why IV access, fluids and a vasopressor (such as phenylephrine or ephedrine) must be ready before induction.

2.3.4 Blood volume

Patient	Approximate blood volume
Premature neonate	90–100 mL/kg
Term neonate	80–90 mL/kg
Infant	75–80 mL/kg
Adult male	70 mL/kg
Adult female	65 mL/kg

A 70 kg man therefore has about 4.9 litres of blood. Losing 15% of this volume usually causes few signs; losses above 30% lead to shock (see Chapter 16).

2.3.5 Important blood vessels for access

Vessel	Common use in the OT
Dorsal veins of the hand, cephalic vein, median cubital vein	Peripheral IV cannulation
Internal jugular, subclavian, femoral veins	Central venous catheter (CVP line)
Radial artery	Invasive arterial blood pressure monitoring and blood gas sampling (perform Allen's test first)
Femoral artery	Arterial access in emergencies and angiography
External jugular vein	Emergency peripheral access

2.4 The Nervous System

2.4.1 Organisation

The nervous system is divided into the **central nervous system (CNS)** — the brain and spinal cord — and the **peripheral nervous system (PNS)**, which includes cranial and spinal nerves. The **autonomic nervous system** controls involuntary functions and has two divisions.

Feature	Sympathetic (“fight or flight”)	Parasympathetic (“rest and digest”)
Origin	Thoracolumbar (T1–L2)	Craniosacral (cranial nerves III, VII, IX, X and S2–S4)

Feature	Sympathetic ("fight or flight")	Parasympathetic ("rest and digest")
Heart rate	Increased	Decreased (vagus nerve)
Blood vessels	Constricted	Little effect
Bronchi	Dilated	Constricted
Pupils	Dilated	Constricted
Gut	Reduced activity	Increased activity and secretions
Main neurotransmitter	Noradrenaline (at end-organ)	Acetylcholine
Drugs that mimic	Adrenaline, ephedrine, phenylephrine	Neostigmine (indirectly)
Drugs that block	Beta-blockers (e.g. metoprolol)	Atropine, glycopyrrrolate

Clinical Pearl — Vagal reflexes in the OT: Traction on the eye muscles (oculocardiac reflex), on the peritoneum, or stretching of the anal sphincter can trigger a sudden vagal slowing of the heart. The surgeon should stop the stimulus immediately, and **atropine** should be kept ready.

2.4.2 The vertebral column

The vertebral column has **33 vertebrae**: 7 cervical, 12 thoracic, 5 lumbar, 5 sacral (fused to form the sacrum) and 4 coccygeal (fused to form the coccyx).

When a spinal or epidural needle is inserted in the midline, it passes through the following layers:

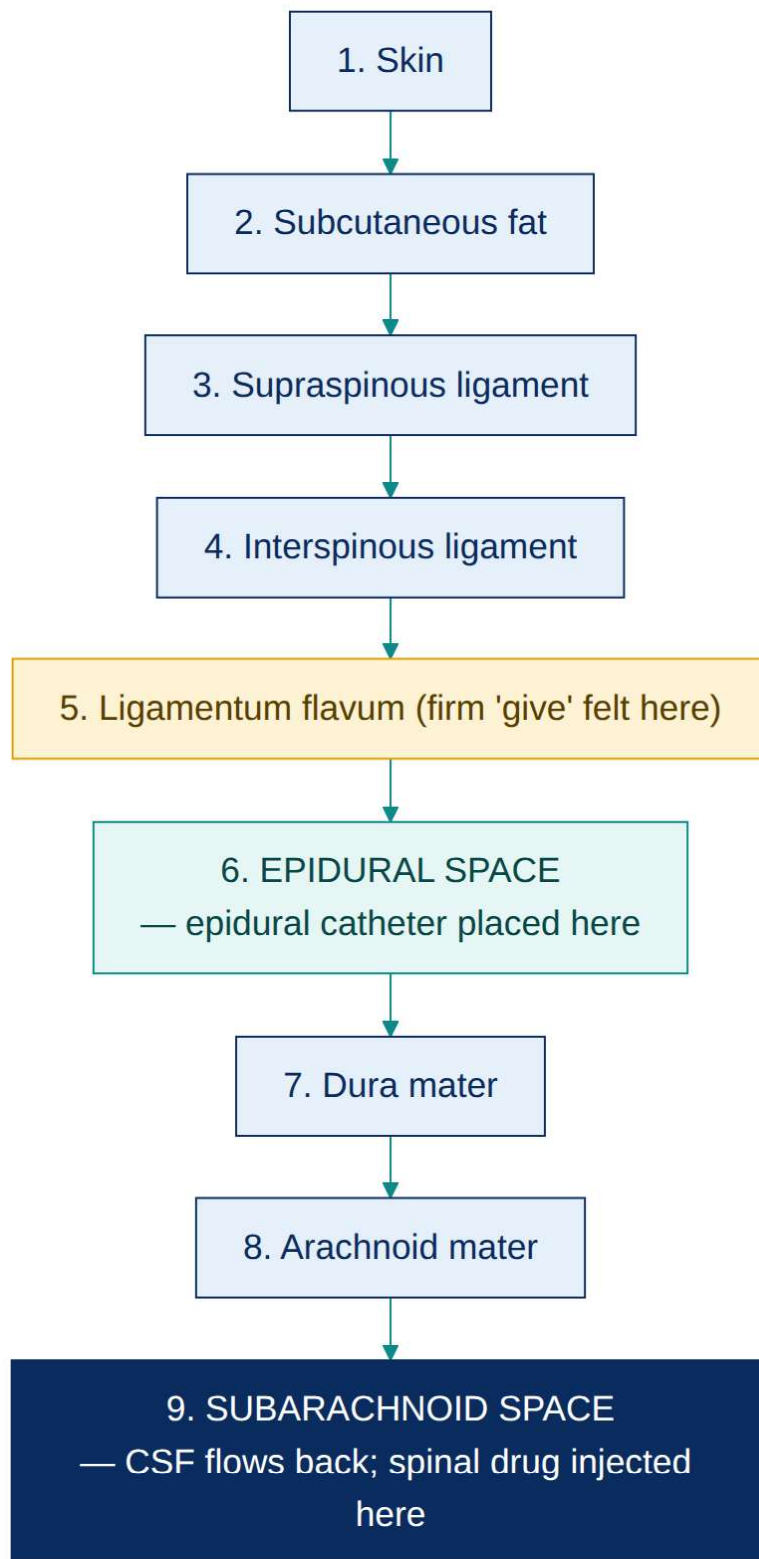


Figure 2.4 — Layers crossed by a spinal needle (midline approach)

Structure	Level in adults	Importance
End of spinal cord (conus medullaris)	L1-L2	Spinal needles are inserted below L2 (usually L3-L4 or L4-L5) to avoid injuring the cord
End of dural sac	S2	Below this, only the epidural (caudal) space remains
Line joining the iliac crests (Tuffier's line)	L4 spinous process or L4-L5 interspace	Landmark for spinal anesthesia
Cerebrospinal fluid (CSF)	Total ~150 mL; produced ~500 mL/day	Clear CSF confirms correct needle position in spinal anesthesia

In children, the spinal cord ends lower (about L3 at birth), so spinal needles are placed at L4-L5 or L5-S1.

2.4.3 Dermatomes

A **dermatome** is an area of skin supplied by a single spinal nerve. Dermatome levels are used to test the height of a spinal or epidural block.

Landmark	Dermatome level	Block needed for
Nipple line	T4	Caesarean section, upper abdominal surgery
Xiphisternum	T6	Upper abdominal surgery
Umbilicus	T10	Lower abdominal surgery, hernia, TURP
Inguinal ligament	T12 / L1	Lower limb surgery
Knee	L3-L4	Surgery below the knee
Perineum	S2-S5	Perineal and anal surgery (saddle block)

Remember — “T4 teat pore, T10 belly button”: T4 at the

nipples and T10 at the umbilicus are the two most commonly tested dermatomes.

2.4.4 The neuromuscular junction

The neuromuscular junction is where a motor nerve meets a skeletal muscle fibre. When a nerve impulse arrives, the nerve ending releases **acetylcholine (ACh)**, which crosses a narrow gap (the synaptic cleft) and binds to **nicotinic receptors** on the muscle. This causes the muscle to contract. The enzyme **acetylcholinesterase** then quickly breaks down ACh so the muscle can relax.

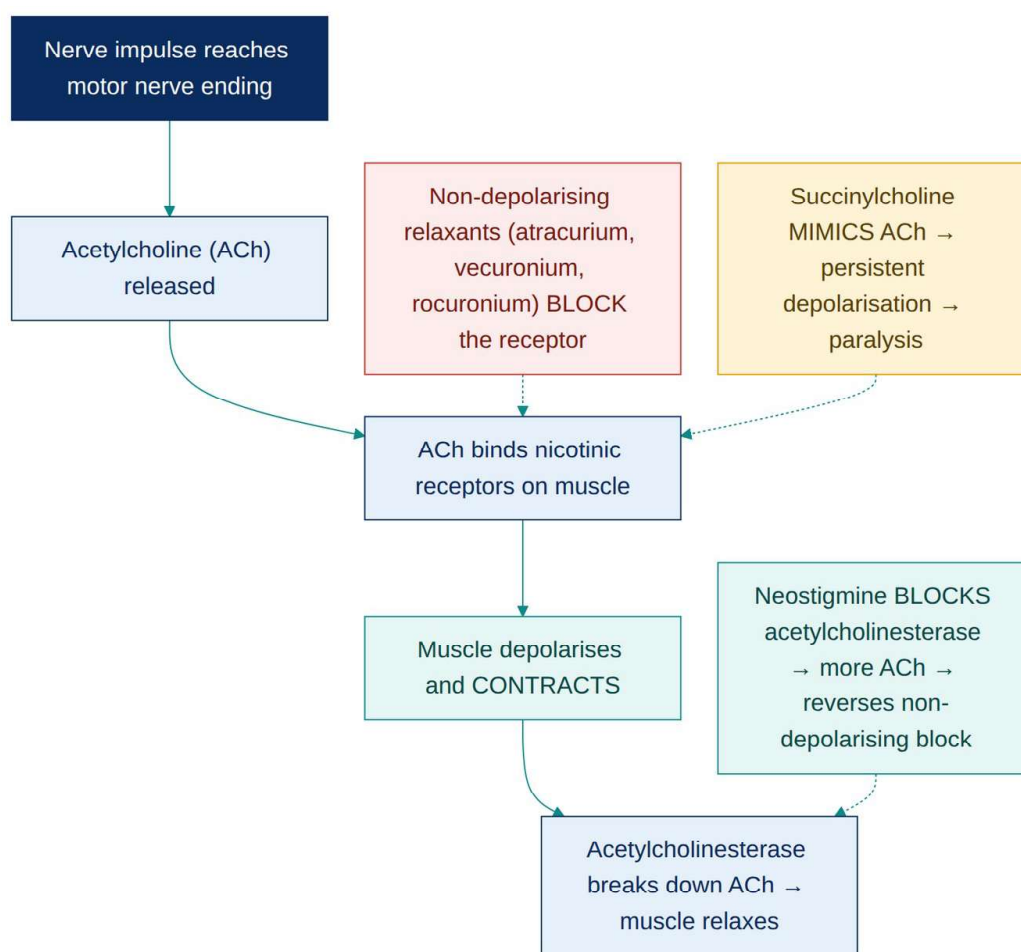


Figure 2.5 — Events at the neuromuscular junction and where drugs act

The pharmacology of these drugs is covered in Chapter 13.

2.5 Body Fluids and Electrolytes

About **60% of an adult man's body weight** is water (about 55% in women, and 75–80% in newborns). This water is distributed between compartments.

Compartment	Proportion of total body water	Volume in a 70 kg man
Intracellular fluid (ICF)	$\frac{2}{3}$	~28 L
Extracellular fluid (ECF)	$\frac{1}{3}$	~14 L
— Interstitial fluid	$\frac{3}{4}$ of ECF	~10.5 L
— Plasma (intravascular)	$\frac{1}{4}$ of ECF	~3.5 L

Electrolyte	Main compartment	Normal plasma value
Sodium (Na^+)	Extracellular	135–145 mEq/L
Potassium (K^+)	Intracellular	3.5–5.0 mEq/L
Chloride (Cl^-)	Extracellular	98–106 mEq/L
Calcium (total)	Extracellular	8.5–10.5 mg/dL
Bicarbonate (HCO_3^-)	Extracellular	22–26 mEq/L
Glucose (fasting)	—	70–100 mg/dL

Safety Alert — Potassium: Both high and low potassium can cause fatal arrhythmias. Concentrated potassium chloride must **never** be given as a direct IV push. It must be diluted and given slowly through an infusion. Many hospitals store concentrated KCl separately as a “high-alert medication”.

2.6 Blood and Haemostasis

Component	Normal adult value	Function
Haemoglobin	Men 13–17 g/dL; women 12–15 g/dL	Oxygen carriage
Haematocrit (PCV)	Men 40–50%; women 36–46%	Proportion of blood made up of red cells
White cell count	4,000–11,000 /mm ³	Defence against infection
Platelet count	1.5–4.5 lakh /mm ³	Initial plug in

Component	Normal adult value	Function
		haemostasis
Prothrombin time (PT) / INR	PT 11–13.5 s; INR 0.8–1.2	Extrinsic clotting pathway; warfarin monitoring
aPTT	25–35 s	Intrinsic pathway; heparin monitoring
Bleeding time	2–7 min	Platelet function
Clotting time	5–10 min	Overall clotting

Haemostasis (the stopping of bleeding) occurs in three overlapping steps: (1) **vascular spasm** — the injured vessel constricts; (2) **platelet plug** formation; (3) **coagulation** — a cascade of clotting factors converts fibrinogen into a fibrin mesh that stabilises the clot. Surgical haemostasis supports these natural steps with pressure, clamps, ligatures, diathermy and haemostatic agents.

2.6.1 Blood groups

Blood group	Antigen on red cells	Antibodies in plasma	Can receive red cells from
A	A	Anti-B	A, O
B	B	Anti-A	B, O
AB	A and B	None	A, B, AB, O (universal recipient)
O	None	Anti-A and anti-B	O only (universal donor)

The Rh (D) factor is also tested: Rh-negative patients should receive Rh-negative blood. Transfusion practice is covered in Chapter 16.

2.7 Temperature Regulation

Normal core body temperature is about **36.5–37.5 °C**. It is controlled by the **hypothalamus**, which balances heat production (metabolism, shivering) against heat loss.

Patients lose heat rapidly in the operation theatre by four mechanisms:

Mechanism	How heat is lost	Example in the OT	Prevention
Radiation (largest, ~40%)	Heat emitted to cooler surroundings	Exposed skin in a cold room	Cover the patient; warm the room for children
Convection	Heat carried away by moving air	Laminar air-flow systems	Forced-air warming blankets
Conduction	Direct contact with cold surfaces	Cold OT table; cold IV fluids	Warming mattress; fluid warmers
Evaporation	Loss of heat as water evaporates	Open abdomen; wet skin-prep solutions; dry anesthetic gases	Minimise exposure; heat-moisture exchanger (HME)

General anesthesia impairs the body's defences against cold: it causes vasodilation (moving warm core blood to the cold periphery — **redistribution**), abolishes shivering, and lowers the metabolic rate. Most of the fall in core temperature occurs in the **first hour** of anesthesia.

Safety Alert — Perioperative hypothermia (core temperature < 36 °C) increases wound infection, blood loss, cardiac complications, drug duration and recovery time. Neonates, the elderly, burns patients and long operations are at the highest risk. Measure temperature in every case longer than 30 minutes and actively warm the patient.

2.8 Other Systems in Brief

System	Points of relevance to the OT technologist
Liver	Metabolises most anesthetic drugs; liver disease prolongs drug effects and impairs clotting
Kidney	Excretes drugs and maintains fluid and electrolyte balance; urine output ≥ 0.5 mL/kg/h indicates adequate perfusion
Gastrointestinal tract	Risk of aspiration of stomach contents during anesthesia — the reason for fasting and rapid

System	Points of relevance to the OT technologist
	sequence induction
Skin	Pressure areas (heels, sacrum, occiput, elbows) are at risk during long operations; skin is the first barrier to infection
Musculoskeletal	Joints must be kept in a neutral position to avoid nerve and ligament injury during positioning
Endocrine	Diabetes (glucose control), thyroid disorders and steroid-dependent patients need special perioperative care

Chapter Summary

- The narrowest part of the adult airway is the glottis; in young children it is the cricoid ring.
- The right main bronchus is shorter, wider and more vertical — tubes and foreign bodies go right.
- FRC is the oxygen store during apnoea; pre-oxygenation fills it with oxygen.
- Blood pressure = cardiac output $\times$ vascular resistance; nearly all anesthetic techniques lower it.
- The spinal cord ends at L1-L2 in adults, so spinal needles are inserted at L3-L4 or below.
- T4 = nipples, T10 = umbilicus.
- Acetylcholine at the neuromuscular junction is the target of muscle relaxants and reversal agents.
- About 60% of body weight is water; two-thirds is intracellular.
- Radiation is the main route of heat loss; most cooling occurs in the first hour of anesthesia.

Review Questions

Multiple Choice Questions

1. The narrowest part of the airway in an infant is the:

- a) Glottis b) Cricoid ring c) Trachea d) Nasopharynx
2. The carina is located at the level of:
 - a) C6 b) T2 c) T4-T5 d) T8
3. Functional residual capacity equals:
 - a) $TV + IRV$ b) $ERV + RV$ c) $IRV + TV + ERV$ d) $VC + RV$
4. In adults, the spinal cord usually ends at:
 - a) T12 b) L1-L2 c) L4-L5 d) S2
5. The dermatome at the level of the umbilicus is:
 - a) T4 b) T6 c) T10 d) L1
6. The neurotransmitter at the neuromuscular junction is:
 - a) Noradrenaline b) Dopamine c) Acetylcholine d) Serotonin
7. The largest source of heat loss in the anaesthetised patient is:
 - a) Conduction b) Convection c) Evaporation d) Radiation
8. Cricoid pressure is applied on the:
 - a) Thyroid cartilage b) Cricoid cartilage c) Hyoid bone d) Tracheal rings
9. The universal donor for red cells is blood group:
 - a) A b) B c) AB d) O
10. The natural pacemaker of the heart is the:
 - a) AV node b) Bundle of His c) SA node d) Purkinje fibres

Answers: 1-b, 2-c, 3-b, 4-b, 5-c, 6-c, 7-d, 8-b, 9-d, 10-c

Short Answer Questions

1. Compare the infant and adult airway.

2. Why does an endotracheal tube tend to enter the right main bronchus?
3. Draw and label the lung volumes and capacities.
4. List, in order, the layers crossed by a spinal needle.
5. Explain the four mechanisms of heat loss in the OT and how each can be prevented.

Long Answer Questions

1. Describe the anatomy of the larynx and its importance in airway management.
2. Describe the physiology of the neuromuscular junction and explain how depolarising and non-depolarising muscle relaxants act.
3. Discuss the causes, effects and prevention of perioperative hypothermia.

Chapter 3: The Operation Theatre Complex — Design, Zones & Environment

Learning Objectives — After studying this chapter, you should be able to:

- Describe the ideal location and layout of an operation theatre complex.
- Explain the four zones of the OT and the rules of movement between them.
- List the rooms and support areas in an OT complex and their functions.
- State the recommended standards for ventilation, air changes, temperature, humidity and lighting.
- Explain the principles of laminar air flow and HEPA filtration.
- Describe the medical gas pipeline system and colour codes.
- Outline the daily, between-case and terminal cleaning routines.

Key Terms: Protective zone · Clean zone · Sterile zone · Disposal zone · Positive pressure · Air changes per hour (ACH) · HEPA filter · Laminar air flow · Medical gas pipeline · Terminal cleaning · Modular OT

3.1 Introduction

The operation theatre complex is a specially designed unit in which the environment itself is a tool of infection control. The walls, floors, doors, air supply, lighting and the direction in which people and materials move are all planned so that micro-organisms are kept away from the open surgical wound. A well-designed OT also makes work efficient, protects staff from hazards, and allows rapid response to emergencies.

3.2 Location of the OT Complex

The OT complex should be:

- **Isolated from general hospital traffic** — usually on a separate floor or a separate wing, away from the main corridors and the hospital entrance.
- **Close to supporting departments** — the surgical wards, ICU, emergency department, blood bank, radiology, and CSSD (ideally linked directly by a clean lift or corridor).
- **Away from sources of contamination** — kitchens, laundries, waste storage areas and outpatient departments.
- **Above ground level** when possible, to reduce dust and flooding, but with access for trolley and stretcher lifts.
- **Planned for expansion** and with separate entries for staff, patients, clean supplies and waste removal.

3.3 The Four Zones of the OT

The OT complex is divided into zones of increasing cleanliness. The basic principle is that **movement should always be from a less clean zone to a cleaner zone only after the correct preparation**, and dirty materials should leave by a route that does not cross clean areas.

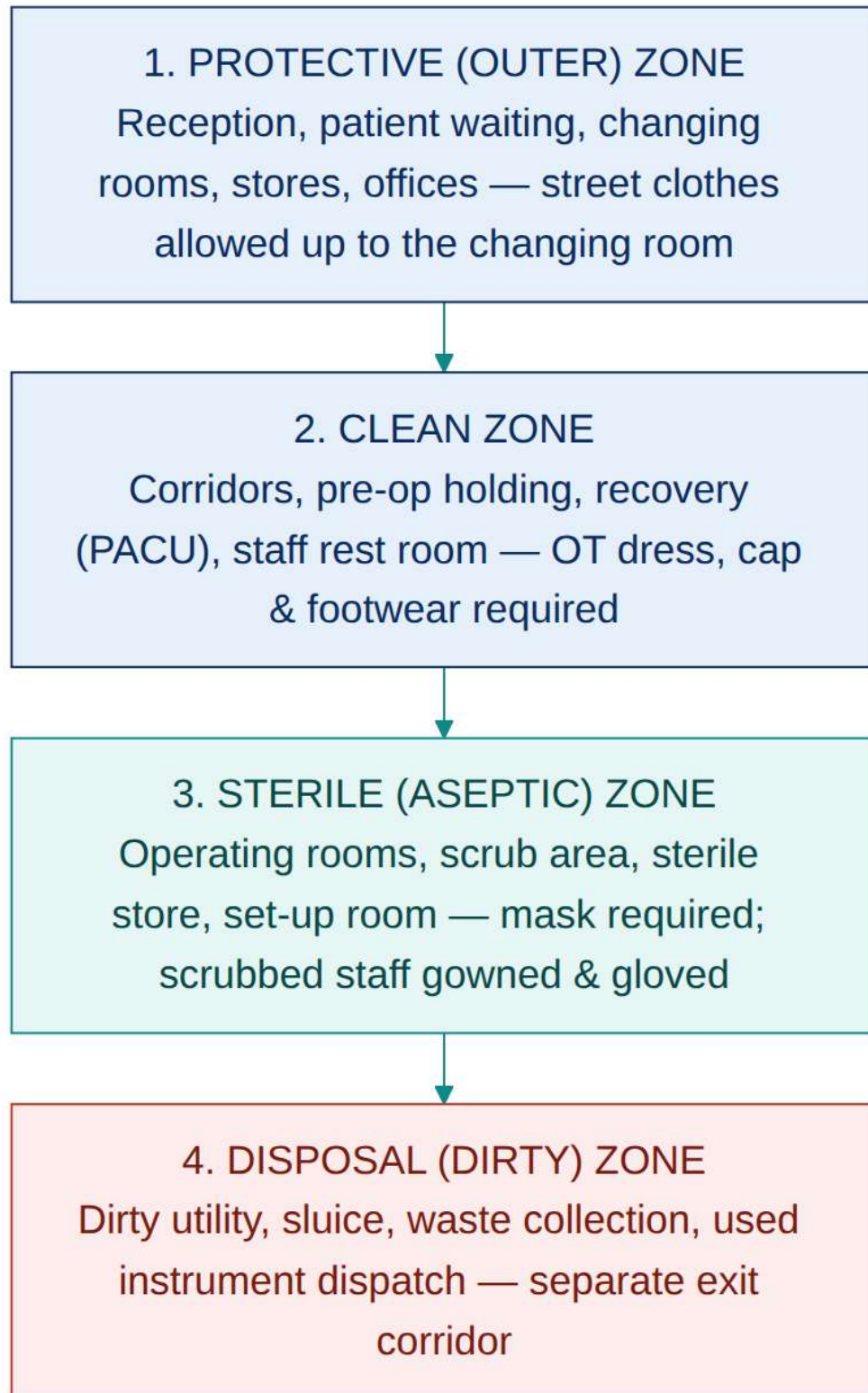


Figure 3.1 — The four zones of the operation theatre complex

Zone	Areas included	Dress code	Key rules
Protective (outer)	Reception, waiting, changing rooms, administrative offices, stores for sealed supplies	Street clothes (until changing room)	Transfer of patients from ward trolley to OT trolley; visitors restricted beyond this point
Clean	Internal corridors, pre-operative holding, PACU, staff lounge, anesthesia store	OT scrub suit, cap, OT footwear	Mask required near sterile rooms; no outside trolleys
Sterile (aseptic)	Operating rooms, scrub bays, sterile supply store, lay-up/preparation room	Full OT attire + mask; scrubbed team gowned and gloved	Doors kept closed; traffic minimised; positive pressure
Disposal (dirty)	Dirty utility, sluice room, waste holding, instrument decontamination/dispatch	OT attire + protective apron, heavy-duty gloves	One-way exit; must not cross the clean corridor

Remember — “Clean in, dirty out”: People, patients and sterile supplies flow *inward* through progressively cleaner zones; used instruments, linen and waste flow *outward* through the disposal zone. The two streams should never meet.

3.4 Traffic Flow Patterns

There are two main design approaches to traffic flow.

Design	Description	Advantages	Disadvantages
Single corridor	One corridor used for clean and dirty traffic, with timing and	Less space and cost; common in older and smaller hospitals	Higher risk of cross-contamination; relies on discipline

Design	Description	Advantages	Disadvantages
	covered transport to separate them		
Double (dual) corridor	A central clean (sterile supply) corridor and a separate peripheral dirty (return) corridor	Clear separation of clean and dirty traffic; preferred in modern hospitals	More space and higher construction cost

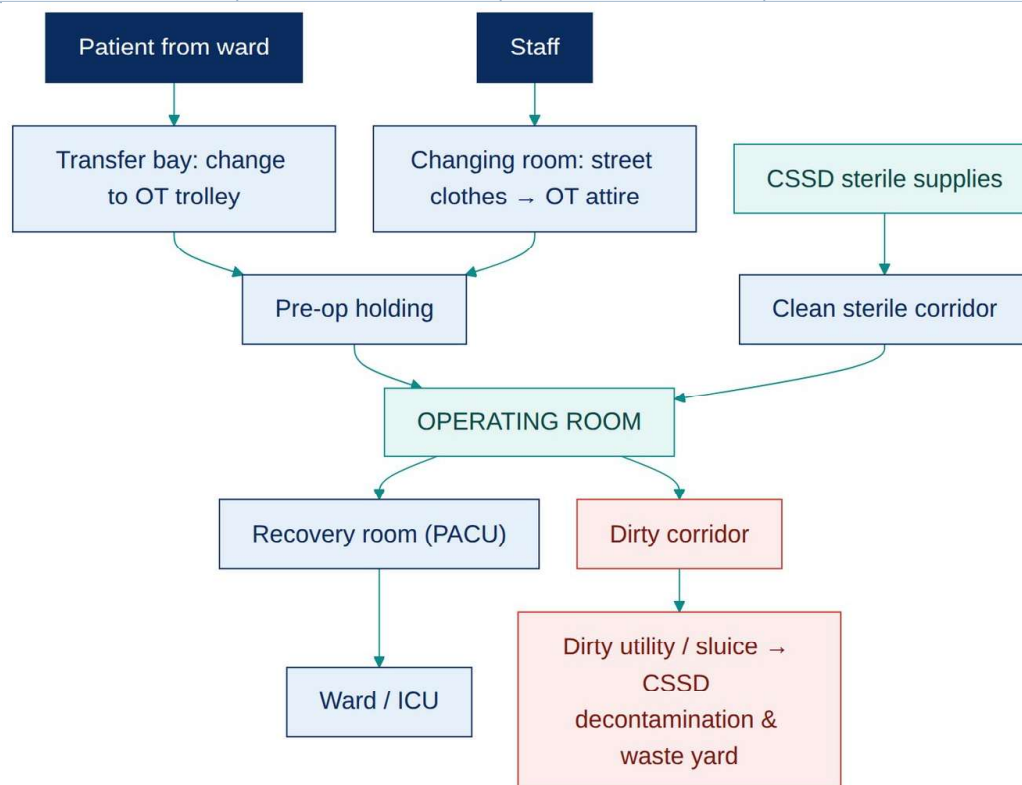


Figure 3.2 — Flow of patients, staff, supplies and waste in a double-corridor OT

3.5 Rooms and Support Areas of the OT Complex

Area	Function
Reception and transfer bay	Receiving patients; verification of documents; transfer from ward trolley to clean OT trolley
Changing rooms (male and	Change from street clothes to

Area	Function
female)	OT attire; lockers; toilets outside the restricted area
Pre-operative holding area	Final checks, IV cannulation, pre-medication, regional blocks
Induction room (in some OTs)	Induction of anesthesia away from the noise of instrument preparation — common in UK-style theatres
Scrub room / scrub bay	Elbow- or sensor-operated taps, antiseptic dispensers, clock; next to each OR
Operating room	The main room where surgery is performed
Sterile store / lay-up room	Storage of sterile packs and instrument sets; trolley preparation
Anesthesia store and workroom	Storage of drugs, airway equipment, circuits; cleaning of anesthesia equipment
Recovery room (PACU)	Monitoring of patients recovering from anesthesia; ideally 1.5–2 beds per operating room
Dirty utility / sluice room	Initial rinsing of instruments, emptying of body fluids, waste segregation
Equipment store	C-arm, microscope, laparoscopy towers, positioning accessories
Doctors' and staff rest rooms	Rest and refreshment within the clean zone
Frozen section / pathology room (optional)	Rapid intraoperative tissue diagnosis
Plant room / AHU room	Air-handling units, filters; usually outside the restricted area for maintenance access

3.6 Design of the Operating Room

3.6.1 Size and shape

Type of OR	Recommended floor area (approximate)
Minor / day-care procedures	20–25 m ²

Type of OR	Recommended floor area (approximate)
General surgery OR	36–42 m ² (e.g. 6.5 m × 6.5 m)
Major specialty OR (orthopaedic, neuro, cardiac, robotic, hybrid)	55–65 m ² or more

A **square or rectangular** room is preferred, with a ceiling height of at least **3 metres** to accommodate ceiling-mounted lights, pendants and laminar flow diffusers. Corners should be **coved (rounded)** so they can be cleaned easily.

3.6.2 Surfaces

Surface	Requirements
Walls	Smooth, seamless, non-porous, washable, resistant to chemicals; pre-painted steel panels or antibacterial laminates in modular OTs; light, non-reflective colours (pastel green or blue reduce eye fatigue)
Floor	Seamless, non-slip, anti-static (conductive) vinyl or epoxy; coved junction with walls; no drains in the operating room
Ceiling	Seamless, sealed, flush-mounted fittings; no false ceiling tiles that shed dust
Doors	Wide (at least 1.2–1.5 m) for trolleys; sliding hermetically sealed doors preferred; vision panel; kept closed during surgery
Windows	Preferably none in modern ORs; if present, sealed and double-glazed

3.6.3 The modular OT

Many new hospitals install **modular operation theatres** — pre-fabricated rooms of steel or glass panels that are assembled on site. Modular OTs offer seamless joints, built-in pendants, integrated control panels (displaying

temperature, humidity, pressure, time and gas alarms), laminar air flow, and faster construction.

3.7 Ventilation and Air Quality

Airborne bacteria, carried mostly on skin scales shed by people in the room, are an important source of surgical site infection. The ventilation system is designed to dilute and remove these particles and to prevent contaminated air from entering the OR.

3.7.1 Standards for the operating room environment

Parameter	Recommended value*	Reason
Air changes per hour (ACH)	At least 20 total ACH (of which at least 4 are fresh outside air); ultra-clean ORs often more	Dilution and removal of airborne contaminants
Pressure	Positive relative to corridors (about +2.5 Pa or more)	Air flows <i>out</i> of the OR, never in from dirtier areas
Filtration	Pre-filter → fine filter → HEPA terminal filter ($\geq 99.97\%$ efficiency for $0.3\ \mu\text{m}$ particles)	Removes bacteria-carrying particles
Temperature	21–24 °C (higher for neonates and burns; as low as 18–20 °C for some cardiac/ortho cases)	Staff comfort and patient normothermia
Relative humidity	40–60% (some standards 20–60%)	Too dry → static electricity and dust; too humid → microbial growth and condensation
General illumination	Around 1000 lux in the room; 40,000–160,000 lux at the surgical site from the operating light	Visibility without glare

*Exact values vary between national guidelines (for example, NABH, HTM 03-01, ASHRAE 170). Always follow your institution's standard.

Clinical Pearl — Keep the doors closed: Every time an OR door opens, the positive pressure falls and turbulent air from the corridor can enter. Studies have found OR doors opened many dozens of times during a single joint replacement. Plan supplies in advance so the circulator does not need to leave the room repeatedly.

3.7.2 Types of air flow

Type	Description	Typical use
Conventional (turbulent / mixing) ventilation	Filtered air enters through ceiling diffusers and mixes with room air; exhaust near the floor	General surgical ORs
Laminar (unidirectional) air flow (LAF)	A large ceiling canopy delivers HEPA-filtered air downward in parallel streams at uniform velocity (about 0.3 m/s), sweeping particles away from the operating table	Joint replacement, spine and neurosurgery, transplant, cardiac surgery

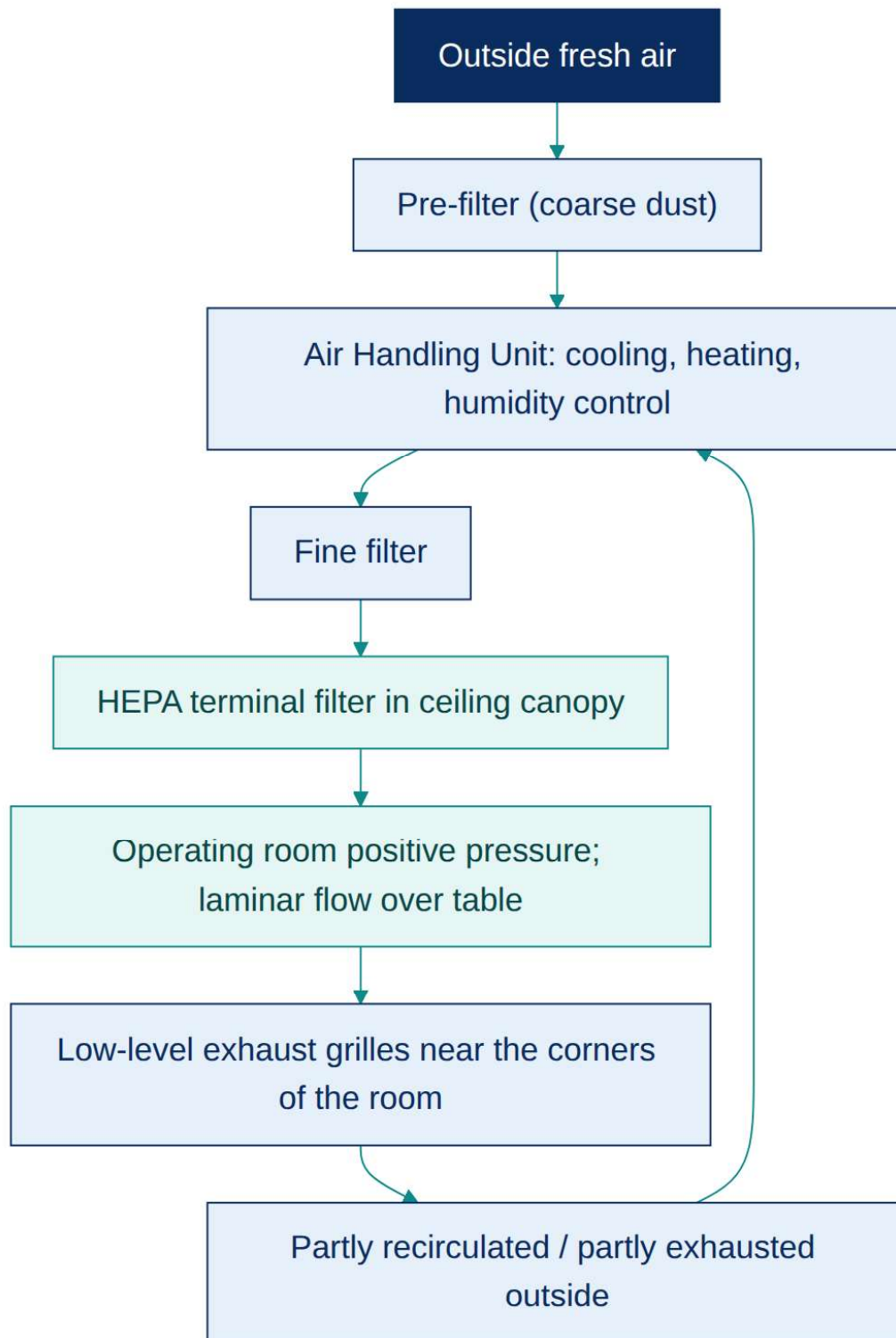


Figure 3.3 — Air handling in the operating room

Safety Alert — Do not block the airflow: Placing tall equipment, stacked linen or personnel between the laminar-flow

canopy and the surgical wound disrupts the clean air stream. Never place trolleys or equipment in front of the low-level exhaust grilles.

3.8 Lighting

The **operating light (surgical light)** must provide intense, shadow-free, cool illumination of the wound.

Requirement	Explanation
Intensity	Adjustable; typically 40,000–160,000 lux at the centre of the light field
Shadow reduction	Multiple light sources (LED clusters or multiple reflectors) so that a head or hand does not cast a shadow
Colour rendering	Colour temperature about 3,500–5,000 K with a high colour-rendering index so tissues look natural
Low heat	Filtered or LED lights minimise heat at the wound and on the surgeon's head
Manoeuvrability	Ceiling-mounted on balanced arms; sterile detachable handles allow the surgeon to adjust the light
Back-up	Connected to emergency power with automatic switchover

LED operating lights have largely replaced halogen lights because they produce less heat, last longer and use less power.

3.9 Electrical Safety and Power Supply

- The OT must be connected to an **uninterruptible power supply (UPS)** and a **diesel generator** with automatic changeover within seconds.
- Life-support equipment (anesthesia machine, ventilator, monitors, operating lights) must be connected to the emergency supply.

- **Isolated power systems** and **line isolation monitors** are used in many ORs to reduce the risk of electric shock.
- All equipment should be **earthed (grounded)** and checked regularly by the biomedical engineering department.
- Floors are anti-static to prevent static sparks (important historically with flammable anesthetics, and still relevant to electronic equipment).
- Multi-socket extension boards and tangled cables on the floor must be avoided.

3.10 Medical Gas Pipeline System (MGPS)

Modern hospitals supply oxygen, nitrous oxide, medical air, carbon dioxide and vacuum through a central **pipeline system** to wall or pendant outlets in each OR. Pipeline outlets are **gas-specific** — each gas has a uniquely shaped connector (a non-interchangeable system) so that a hose for one gas cannot be plugged into the outlet for another.

Gas	Pipeline pressure (approx.)	Colour of pipeline / outlet (ISO 32 standard, used in India)	Cylinder colour (India)
Oxygen (O₂)	4 bar ($\approx$ 60 psi)	White	Black body with white shoulder
Nitrous oxide (N₂O)	4 bar	Blue	Blue (French blue)
Medical air	4 bar	Black-and-white	Grey body with black-and-white shoulder
Surgical air (for power tools)	7 bar	Black-and-white	—
Carbon dioxide (CO₂)	— (cylinder, via insufflator)	Grey	Grey
Vacuum (suction)	Negative pressure (about -400 to	Yellow	—

Gas	Pipeline pressure (approx.)	Colour of pipeline / outlet (ISO 32 standard, used in India)	Cylinder colour (India)
	–500 mmHg)		
Entonox (50% O₂ / 50% N₂O)	—	Blue/white	Blue body with blue-and-white shoulder

Colour codes differ between countries (for example, in the USA oxygen is green). Always read the label on a cylinder or outlet — never rely on colour alone.

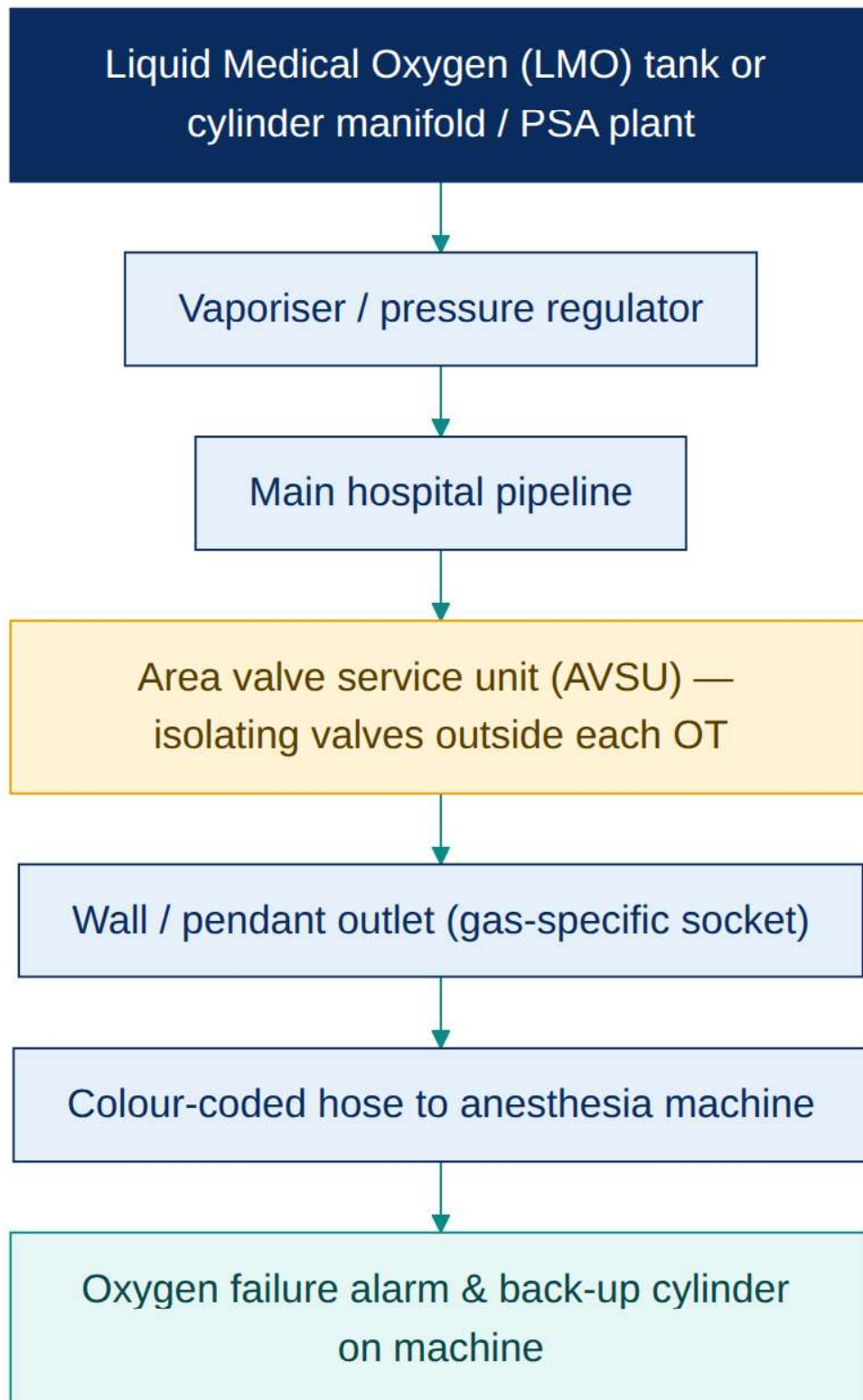


Figure 3.4 — Medical oxygen supply to the operating room

Safety Alert — Pipeline crossover: Deaths have occurred worldwide when nitrous oxide was accidentally supplied through the oxygen pipeline after construction or repair work. After any work on the pipeline, the gas at each outlet must be tested with an **oxygen analyser** before use. The anesthesia machine's oxygen analyser must be calibrated and alarms switched on for every case.

3.11 Furniture and Fixed Equipment in the OR

Item	Purpose
Operating table	Supports and positions the patient (Chapter 7)
Operating lights	Illumination of the surgical field
Anesthesia workstation	Delivery of gases and ventilation (Chapter 10)
Multiparameter monitor	Patient monitoring (Chapter 12)
Ceiling pendants	Gas outlets, power sockets, data points; keep cables off the floor
Instrument trolleys and Mayo stand	Sterile set-up of instruments
Electrosurgical unit	Cutting and coagulation (Chapter 7)
Suction apparatus	Wall suction plus a portable back-up unit
X-ray viewing screen / PACS display	Viewing imaging during surgery
Wall clock with elapsed-time counter	Recording times of incision and procedures
Kick buckets, linen hampers, waste bins	Swab collection and waste segregation
Emergency trolley and defibrillator	Within immediate reach of every OR

3.12 Cleaning of the Operation Theatre

Cleaning removes dust, organic matter and micro-organisms from surfaces. It follows a set routine.

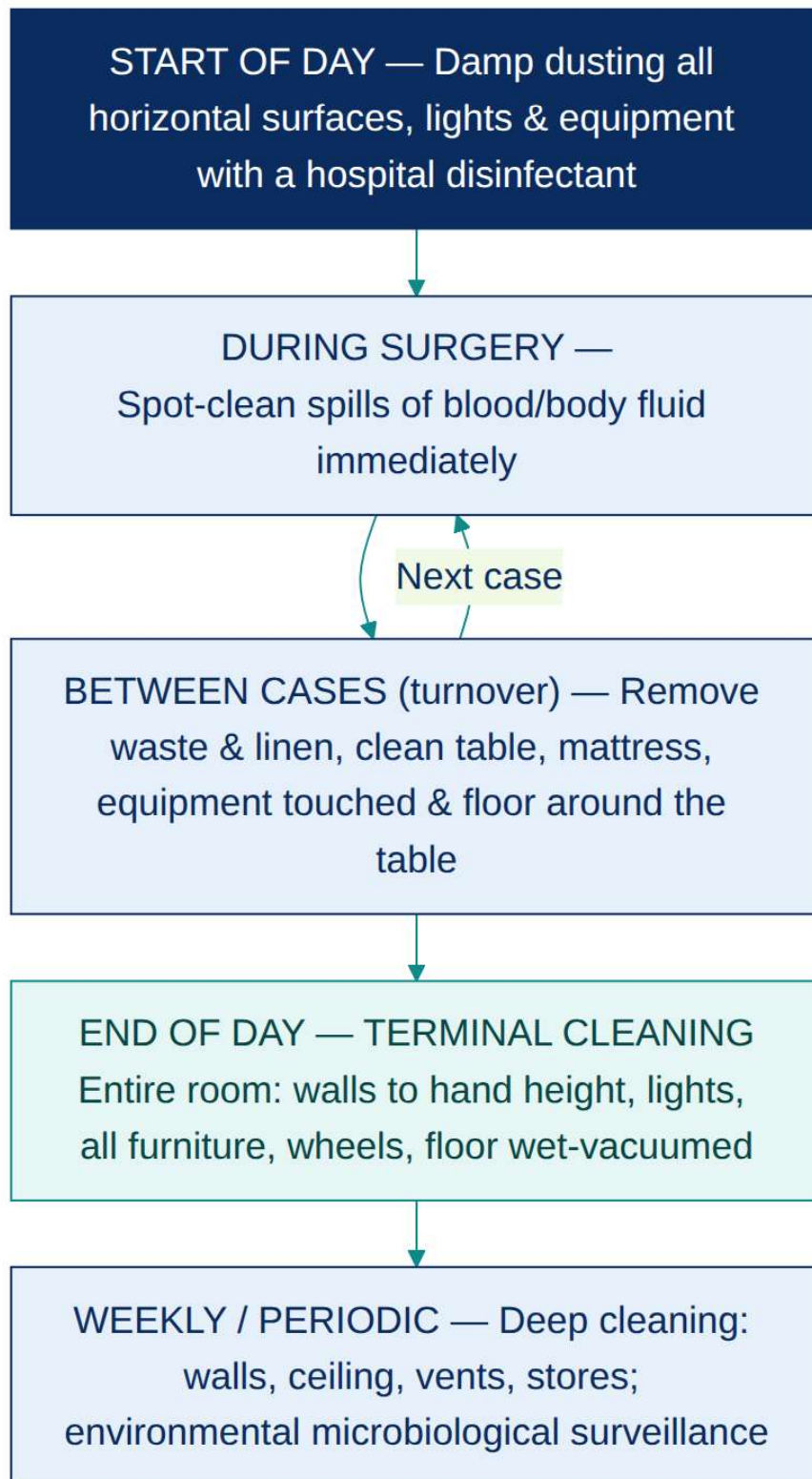


Figure 3.5 — The OT cleaning cycle

Type of cleaning	When	What is done
Damp dusting	Before the first case each day	Wipe all horizontal surfaces, lights, table, trolleys and equipment with a lint-free cloth moistened with disinfectant
Spot cleaning	Immediately when a spill occurs	Cover blood spill with absorbent material, apply sodium hypochlorite (1% = 10,000 ppm available chlorine for blood spills, or as per hospital policy), leave for the recommended contact time, wipe, then clean
Between-case (turnover) cleaning	After every procedure	Remove all waste, sharps, linen; clean the OR table, mattress, arm boards, straps, equipment in contact with the patient, and floor in the area of activity (about 1-1.5 m around the table)
Terminal cleaning	At the end of the day's list	Clean all surfaces top to bottom, including lights, pendants, walls to hand height, doors and handles, furniture, wheels and castors; wet-vacuum or mop the entire floor
Deep / weekly cleaning	Weekly or as scheduled	Walls and ceiling, air grilles, storage shelves, scrub sinks; environmental sampling as per infection control policy

Clinical Pearl — Cleaning order: Always clean from **clean to dirty** and from **top to bottom** — lights and high surfaces first, then furniture, then floor. Use separate colour-coded mops and cloths for the OR and for toilets and sluice rooms.

3.12.1 Fumigation — current thinking

In the past, many Indian hospitals routinely fumigated OTs with formalin vapour. Formaldehyde is a **toxic and carcinogenic** chemical, and routine fogging is now discouraged by modern infection control guidance. Thorough mechanical cleaning with approved surface disinfectants (for example, hydrogen peroxide–silver nitrate combinations, quaternary ammonium compounds or chlorine-based agents), combined with a properly functioning HEPA air system, is considered more effective. Follow the policy of your hospital’s infection control committee.

3.12.2 Microbiological surveillance

Many hospitals perform periodic **settle-plate** or **air-sampler** tests and surface swabs, especially after construction, maintenance of air-handling units, or an outbreak of infection. Results should be within the limits set by the infection control committee before the OR is used.

3.13 OT Attire and Personal Conduct Rules

Item	Requirement
Scrub suit	Clean, freshly laundered, hospital-provided; shirt tucked in; changed when soiled or wet
Cap / hood	Covers all hair (including beard and sideburns with a hood)
Mask	Covers nose and mouth completely; worn in the sterile zone and whenever sterile items are open; changed between cases or when damp; never left dangling around the neck
Footwear	Dedicated OT shoes or clogs; closed, anti-slip; cleaned regularly

Item	Requirement
Jewellery	Remove rings, watches, bracelets; stud earrings covered by the cap
Nails	Short and natural; no nail polish or artificial nails
Identity badge	Worn, cleaned regularly
Personal items	Mobile phones kept to a minimum and cleaned; no food or drink in restricted areas

Safety Alert — Leaving the OT in scrubs: When staff leave the OT complex (for example, to go to the ward or canteen), they should either change or wear a clean cover gown, and they must change into a fresh scrub suit on return according to hospital policy. Wearing theatre clothing outside the hospital is not acceptable.

Chapter Summary

- The OT complex should be isolated from general traffic but close to ICU, emergency, blood bank and CSSD.
- The four zones are protective, clean, sterile and disposal; movement is “clean in, dirty out”.
- A double-corridor design separates clean and dirty traffic.
- Operating rooms are kept at positive pressure with ≥ 20 air changes per hour through HEPA filters, 21–24 °C and 40–60% humidity.
- Laminar air flow is used for implant and ultra-clean surgery.
- Medical gases are delivered through a pipeline with gas-specific outlets; oxygen pipeline colour (ISO) is white; always verify with an oxygen analyser.
- Cleaning routines include damp dusting, spot cleaning, between-case cleaning, terminal cleaning and periodic deep cleaning.
- Routine formalin fumigation is now discouraged; follow the infection control committee’s policy.

Review Questions

Multiple Choice Questions

1. The operating room is maintained at:
a) Negative pressure b) Positive pressure c) Atmospheric pressure d) Variable pressure
2. A HEPA filter removes at least what percentage of particles of 0.3 μm size?
a) 90% b) 95% c) 99.97% d) 100%
3. Recommended relative humidity in the OR is:
a) 10–20% b) 40–60% c) 70–80% d) 90%
4. The scrub room belongs to which zone?
a) Protective b) Clean c) Sterile d) Disposal
5. Cleaning of the entire OR at the end of the day is called:
a) Damp dusting b) Spot cleaning c) Terminal cleaning d) Fumigation
6. According to the ISO colour code used in India, the vacuum pipeline is:
a) White b) Blue c) Yellow d) Black
7. Laminar air flow is most important for:
a) Incision & drainage b) Total joint replacement
c) Circumcision d) Endoscopy
8. The usual pipeline pressure for oxygen is about:
a) 1 bar b) 4 bar c) 10 bar d) 137 bar

Answers: 1-b, 2-c, 3-b, 4-c, 5-c, 6-c, 7-b, 8-b

Short Answer Questions

1. List the four zones of the OT with two examples of areas in each.
2. What is the difference between single and double corridor designs?

3. State the standards for air changes, temperature, humidity and pressure in the OR.
4. What are the requirements of a good operating light?
5. Why must an oxygen analyser be used after pipeline repair work?

Long Answer Questions

1. Describe the planning, zoning and traffic flow of an operation theatre complex with a labelled flowchart.
2. Discuss the ventilation system of the OR, including laminar air flow and HEPA filtration.
3. Describe the cleaning routines of the operation theatre, and discuss the current view on fumigation.

Part II — Operation Theatre Practice

Part II covers the core practical skills of the OT technologist: preventing infection, sterilising instruments and supplies, knowing and caring for surgical instruments and sutures, operating OT equipment, and positioning and protecting the patient on the operating table.

- Chapter 4: Infection Control, Asepsis & Surgical Scrubbing
- Chapter 5: Sterilisation & Disinfection
- Chapter 6: Surgical Instruments, Sutures & Needles
- Chapter 7: OT Equipment & Surgical Technology
- Chapter 8: Surgical Positioning & Patient Safety